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CHAPTER 3

CLASSIFICATION OF COMPUTER NETWORK

SUGGESTED TIME: 3 PERIODS

IN THIS CHAPTER

This chapter explores the classification of computer networks by geographic coverage, the criteria for selecting an appropriate network architecture for an organization, and culminates in an integration activity where learners design a network solution for a realistic Cameroonian scenario.

- Identify and describe the main types of networks (PAN, LAN, CAN, MAN, WAN, VPN) according to their coverage area.
- Compare the characteristics, typical technologies, and use-cases of each network type.
- Explain the role of the Internet as a global WAN and its relationship with private networks.
- Analyse organisational requirements (size, budget, security, scalability, mobility) to choose the most suitable network type.
- Apply a structured decision-making method to recommend a network architecture for a given case study.
- Design a simple network diagram that integrates multiple network types (e.g., LANs linked by a MAN/WAN).

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2. Metropolitan, Wide Area Networks and the Internet
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CHAPTER OBJECTIVES

- Identify and describe the main types of networks (PAN, LAN, CAN, MAN, WAN, VPN) according to their coverage area.
- Compare the characteristics, typical technologies, and use-cases of each network type.
- Explain the role of the Internet as a global WAN and its relationship with private networks.
- Analyse organisational requirements (size, budget, security, scalability, mobility) to choose the most suitable network type.
- Apply a structured decision-making method to recommend a network architecture for a given case study.
- Design a simple network diagram that integrates multiple network types (e.g., LANs linked by a MAN/WAN).
- Evaluate the advantages and limitations of virtual private networks for remote access.
- Communicate network design choices clearly using appropriate terminology and diagrams.

PREREQUISITES

- Basic understanding of data communication concepts (sender, receiver, medium, protocol) from Lower Sixth ICT.

- Familiarity with common network devices (switch, router, access point) and their functions.
- Knowledge of IP addressing fundamentals (IPv4/IPv6, subnet mask, private vs public addresses).
- Awareness of common transmission media (copper, fiber, wireless) and their bandwidth characteristics.

1 PERSONAL, LOCAL AND CAMPUS AREA NETWORKS

In Yaoundé, a small startup shares files between two laptops using a USB stick, while a multinational bank in Douala connects its branches across the country through a dedicated fibre link. Meanwhile, a university student in Buea accesses the campus library database from a café via Wi-Fi, and a government agency monitors traffic cameras nationwide over a secure VPN. These everyday scenes illustrate how networks of vastly different sizes and purposes coexist in Cameroon. In this section we classify networks by their geographic coverage, focusing on the three smallest categories: Personal Area Networks (PAN), Local Area Networks (LAN) and Campus Area Networks (CAN). We will see how each type is built, which technologies dominate, and how to choose the right one for a given situation.

1.1 Personal Area Network (PAN)

A **Personal Area Network (PAN)** is the smallest network you will encounter. It centres on a single person and typically spans less than ten metres. Think of the Bluetooth headset paired with your smartphone, the smartwatch that receives notifications, or the NFC tag you tap to pay for a bus ticket in Douala. All these devices communicate directly with each other or through a master device (usually the phone) without any fixed infrastructure.

DEFINITION **Personal Area Network (PAN)**

A PAN is a network organised around an individual user, covering a radius of roughly 10 m or less. It uses short-range wireless technologies such as Bluetooth (IEEE 802.15.1), Zigbee (IEEE 802.15.4) or Near Field Communication (NFC). Wired variants (USB, FireWire) exist but are less common in modern mobile contexts. No dedicated cabling or central switch is required.

PROPERTY **Typical PAN Technologies**

- Bluetooth – up to 100 m (Class 1), commonly 10 m (Class 2); supports audio streaming, file transfer, IoT sensors.
- Zigbee – low-power, mesh-capable, range 10 m–100 m; used in home automation (smart bulbs, sensors).
- NFC – extremely short range (< 4cm); contactless payments, pairing, access control.

EXEMPLE **Everyday PAN in Cameroon**

A student in Yaoundé uses a smartphone (master) connected via Bluetooth to a wireless headset for online lectures, while a smartwatch receives calendar alerts. When she buys

a snack at the campus canteen, she taps her phone on an NFC terminal. All three links form a single PAN centred on her phone.

1.2 Local Area Network (LAN)

When several computers in a single building need to share resources — printers, files, an Internet connection — a **Local Area Network (LAN)** is the natural choice. A LAN covers a limited geographic area: one room, one floor, or a group of adjacent buildings owned by the same organisation. Unlike a PAN, a LAN relies on a fixed infrastructure: structured cabling (twisted-pair copper or fibre) and/or wireless access points, all converging on switches and a router.

DEFINITION **Local Area Network (LAN)**

A LAN is a high-speed data network that interconnects devices within a single building or a cluster of nearby buildings, typically owned and managed by a single organisation. It provides full-time connectivity with low latency and high bandwidth (commonly 100 Mb/s to 10 Gb/s on wired links, up to 9.6 Gb/s on Wi-Fi 6/6E).

PROPERTY **LAN Technologies and Access Methods**

- Ethernet (IEEE 802.3) – wired LAN standard. In half-duplex mode it uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection); modern full-duplex switched Ethernet eliminates collisions entirely.
- Wi-Fi (IEEE 802.11) – wireless LAN family (802.11a/b/g/n/ac/ax/be). Uses CSMA/CA (Collision Avoidance) because the medium is shared and collisions cannot be detected reliably. A WLAN is a LAN that uses wireless links for the last hop.
- Topology – virtually all modern LANs use a star topology physically (each device to a switch) while the logical topology is a broadcast domain (single subnet unless VLANs are used).

KEY POINTS **Key LAN Characteristics**

- Ownership: private (school, company, home).
- Geographic scope: ≤ 1 km (usually a few hundred metres).
- Speed: 100 Mb/s – 10 Gb/s (wired); 54 Mb/s – 9.6 Gb/s (Wi-Fi).
- Management: centralised via switches, VLANs, DHCP, DNS.

EXEMPLE **School Computer Lab LAN**

A secondary school in Bafoussam equips its computer lab with 30 PCs. Each PC has a NIC (Network Interface Card) connected by Cat 6 cable to a 48-port Gigabit switch. The switch uplinks to a router that provides Internet access and runs a DHCP server. All PCs are in the same VLAN (192.168.10.0/24). Students can print to a network printer and access a local Moodle server.

1.3 Campus Area Network (CAN)

A university, a large hospital, or a corporate headquarters often occupies several buildings spread over a few kilometres. Connecting each building's LAN with a high-speed backbone creates a **Campus Area Network (CAN)**. A CAN is essentially an interconnection of multiple LANs within a limited geographic area, usually owned by a single entity.

DEFINITION **Campus Area Network (CAN)**

A CAN links two or more LANs located in different buildings of a campus (university, hospital, industrial park, government complex) using a high-capacity backbone (often fibre optic). The total span is typically a few kilometres (up to 5 km). The organisation owns and manages the entire infrastructure.

WARNING **CAN vs. MAN – Do Not Confuse!**

A **Metropolitan Area Network (MAN)** covers a whole city (tens of kilometres) and often relies on telecom operator infrastructure (leased lines, MPLS, metro-Ethernet). A CAN is privately owned, shorter, and uses the organisation's own fibre or wireless bridges. In exams, remember: CAN = private campus; MAN = city-wide, often carrier-provided.

EXEMPLE **University of Buea CAN**

The University of Buea runs a CAN linking the Faculty of Engineering, the Library, the Administration block and the Student Centre. Each building has its own LAN (switches, Wi-Fi APs). A 10 Gb/s fibre ring connects the building-level core switches to a central core router. Students authenticate once (eduroam) and roam seamlessly across the campus.

1.4 Comparison of Network Types by Coverage

To consolidate the classification required by the syllabus, the table below summarises the five main categories. MAN and WAN are introduced here for completeness; they are studied in de-

tail in the next section.

PROPERTY **Network Classification by Geographic Scope**

Type	Typical Scope	Ownership	Typical Technologies
PAN	<10 m (personal)	Individual	Bluetooth, Zigbee, NFC, USB
LAN	≤1 km (building)	Organisation	Ethernet, Wi-Fi (802.11)
CAN	≤5 km (campus)	Organisation	Fibre backbone, wireless bridges
MAN	5 km–50 km (city)	Telecom carrier / consortium	Metro-Ethernet, MPLS, leased lines
WAN	>50 km (national/global)	Multiple carriers / Internet	IP/MPLS, submarine cables, satellite

1.5 Common Network Devices

Regardless of the network size, a few devices appear repeatedly:

Device	Role in PAN / LAN / CAN
NIC (Network Interface Card)	Provides the physical interface (RJ-45, Wi-Fi radio, Bluetooth radio) on each end device.
Switch (Layer 2)	Forwards frames based on MAC addresses; creates collision-free domains in LAN/CAN.
Wireless Access Point (AP)	Bridges wireless clients to the wired LAN; implements IEEE 802.11.
Router (Layer 3)	Interconnects different IP subnets (e.g., building LANs in a CAN) and provides gateway to WAN/Internet.
Bluetooth / Zigbee Adapter	USB or built-in radio for PAN peripherals.

1.6 Typical Applications in Cameroon

- **Home office PAN** – freelancer in Douala pairs laptop, mouse, keyboard and phone via Bluetooth; uses NFC for mobile money (MTN MoMo, Orange Money) top-up.
- **School computer lab LAN** – public and private secondary schools (e.g., Lycée Bilingue de Yaoundé) deploy wired LANs for ICT practicals, e-learning platforms, and administrative management.
- **University campus CAN** – University of Yaoundé I, University of Buea, Catholic University of Central Africa interconnect faculty LANs for shared library resources, student information systems, and campus-wide Wi-Fi (eduroam).

- **Health-center CAN** – regional hospitals link laboratory, pharmacy, wards and administration LANs over fibre to enable real-time patient records (e-Health Cameroon pilot projects).

1.7 Method: Setting Up a Simple Wired LAN

The following step-by-step procedure is examinable in practical papers (GCE ICT Paper 2/3). It assumes a small office with 10 PCs, one printer and an Internet router.

METHOD **Deploy a Basic Wired LAN**

1. Plan the IP addressing scheme. Choose a private IPv4 network, e.g., `192.168.20.0/24`. Reserve `.1` for the router (gateway), `.2` for the printer, `.10–.20` for DHCP pool, `.200–.210` for static servers.
2. Run structured cabling. Use Cat 6 UTP, terminate with T568B on both ends. Keep cable runs ≤ 90 m (horizontal) + 10 m (patch cords). Label each outlet (A-01, A-02 …).
3. Install a managed switch. Mount a 24-port Gigabit switch in a locked cabinet. Connect all wall outlets to switch ports. Configure ports as access ports in VLAN 10 (data) or VLAN 20 (voice) if IP phones are used.
4. Configure the router. Assign the LAN interface `192.168.20.1/24`. Enable DHCP server with the pool defined in step 1. Set DNS to `8.8.8.8` and `1.1.1.1` (or internal DNS).
5. Connect the printer. Give it a static IP `192.168.20.2`. Create a DNS entry `printer.lan`.
6. Test connectivity. From each PC: `ping 192.168.20.1`, `ping 192.168.20.2`, browse `http://www.camtelecom.cm`. Verify DHCP lease with `ipconfig /all` (Windows) or `ip a` (Linux).
7. Document. Produce a network diagram (Visio/draw.io), IP allocation table, VLAN map, and switch port-to-outlet mapping. Hand over to IT support.

WORKED EXAMPLE 1 **Worked Exercise: Choosing the Right Network Type**

Statement: A small NGO in Bamenda has three offices in the same building (ground floor, 1st floor, 2nd floor). Each office has 5 laptops. They need to share a printer and a local database server. The NGO also wants the director's smartphone to sync contacts with his laptop via Bluetooth. Classify each required network (PAN, LAN, CAN, MAN, WAN) and justify.
Solution:

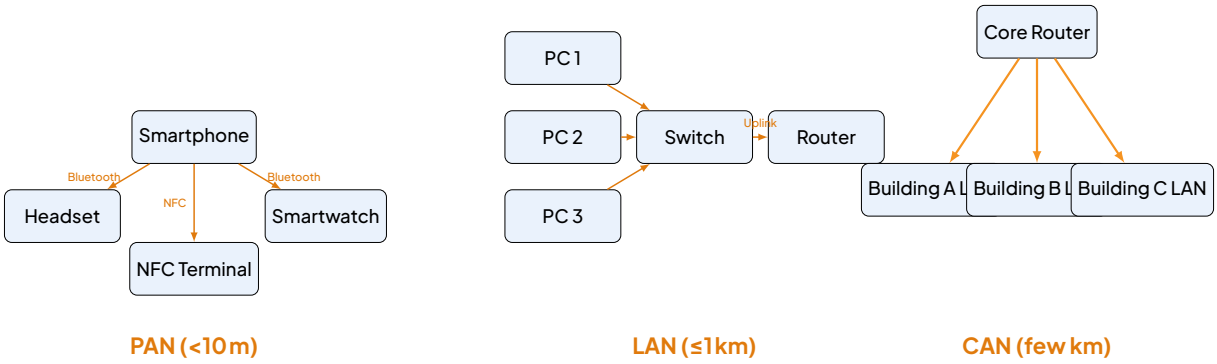
- PAN – Director's smartphone ↔ laptop (Bluetooth, < 10 m). No infrastructure needed.
- LAN – The three offices are in a single building owned by the NGO. A single LAN (star topology, one switch per floor uplinked to a core switch) covers all laptops, printer

and server. Geographic scope < 100 m, private ownership → LAN.

- CAN, MAN, WAN – Not required because there is only one building and no inter-city links.

Answer: PAN for the director's personal devices; LAN for the office network. No CAN, MAN or WAN needed.

1.8 Illustration: PAN, LAN and CAN Overview



Typical scope and devices of a PAN (Bluetooth/NFC), a LAN (PCs–Switch–Router) and a CAN (multiple LANs on a fibre backbone).

1.9 Summary of Key Points

KEY POINTS **What You Must Remember for the Exam**

- PAN: personal, < 10 m, Bluetooth/Zigbee/NFC, no fixed infrastructure.
- LAN: single building or adjacent buildings, private ownership, Ethernet (CSMA/CD or full-duplex) and Wi-Fi (CSMA/CA), star topology, speeds ≥ 100 Mb/s.
- CAN: interconnection of several LANs on a campus (≤ 5 km), private fibre backbone, same owner.
- MAN/WAN: city-wide and beyond, usually carrier-operated (see next section).
- Devices: NIC, switch, wireless AP, router, Bluetooth/Zigbee adapter.
- Typical Cameroonian contexts: home-office PAN, school lab LAN, university CAN.
- Method: plan IP, cable to Cat6 standard, configure switch VLANs, router DHCP/gateway, test, document.

- Trap: never label a city-wide network as a CAN; that is a MAN (often carrier-operated).

DID YOU KNOW? **Did You Know?**

Cameroon's Agence Nationale des Technologies de l'Information et de la Communication (ANTIC) promotes the deployment of campus networks in public universities under the Projet d'Appui à la Gouvernance Numerique. Many campuses now run 10 Gb/s fibre rings, enabling services like e-Learning platforms, eduroam roaming and centralised Library Management Systems — all built on the CAN principles you have just studied.

2 METROPOLITAN, WIDE AREA NETWORKS AND THE INTERNET

In the previous section we studied networks that live inside a limited space: a PAN around one person, a LAN inside one building, a CAN across a campus. But what happens when the Ministry of Secondary Education in Yaoundé must exchange files with its regional delegation in Douala, or when a bank with branches in ten cities must keep all its account databases synchronised? No single LAN can stretch that far. We must climb to the next levels of the classification: the **MAN**, the **WAN**, and ultimately the **Internet**. This section studies these large-scale networks and the technologies — including VPNs — that make them safe to use.

2.1 Metropolitan Area Networks (MAN)

Intuition. Imagine a water distribution network: inside your house you have your own pipes (the LAN), but to bring water from the treatment plant to every house in the city, the city needs a much bigger network of mains. A MAN plays exactly this role for data at the scale of a town or city.

DEFINITION **Metropolitan Area Network (MAN)**

A **Metropolitan Area Network** is a network that spans a city or a metropolitan area, typically from about 5 to 50 km, interconnecting several LANs and CANs located in different sites of the same town. A MAN is usually owned and operated by a telecommunications provider (or a consortium), which sells connectivity to organisations rather than each organisation laying its own cables across public roads.

Typical technologies.

- **Metro Ethernet**: the provider extends Ethernet services across the city over its own fibre

infrastructure, so that two distant sites appear to be plugged into the same switch.

- **Fibre optic ring**: fibre cables are laid in a ring around the city (SDH/SONET or modern DWDM). If the ring is cut at one point, traffic simply travels the other way round — this gives **resilience**.
- **WiMAX** (IEEE 802.16): a wireless metropolitan technology able to cover a whole city from a few base stations; historically used where digging trenches for fibre was too expensive.

Ownership models. Unlike a LAN (owned by the organisation itself), a MAN is usually:

- owned by a **single telecom operator** (e.g. CAMTEL's fibre backbone in Yaoundé and Douala);
- owned by a **municipality or public authority** and rented to operators;
- owned by a **consortium** of large users (universities, banks) sharing the cost.

DID YOU KNOW? **Cameroon: CAMTEL's fibre backbone**

CAMTEL, the historical operator of Cameroon, runs a national fibre-optic backbone linking Yaoundé, Douala, Bafoussam, Bamenda, Garoua and other cities, and operates the landing points of submarine cables (such as SAIL, linking Cameroon to Brazil) at the coast. Within each city, this infrastructure functions as a MAN over which internet service providers, banks and ministries interconnect their sites. The landing of submarine cables at Kribi and Douala is what physically connects Cameroon to the rest of the world.

2.2 Wide Area Networks (WAN)

Intuition. A MAN covers one city. To connect the Yaoundé MAN to the Douala MAN — or a company's head office in Cameroon to a partner in Paris — we need links that cross regions, countries and oceans. That is the job of a WAN.

DEFINITION **Wide Area Network (WAN)**

A **Wide Area Network** is a network that covers large geographic distances — regions, countries or continents — and interconnects multiple LANs, CANs and MANs. Because no single organisation can lay cables across a whole country, a WAN is built from transmission services leased from telecom operators (leased lines, MPLS, satellite, mobile backhaul), and it relies on routers and routing protocols to move data between distant sites.

Typical WAN technologies.

- **Leased line**: a dedicated point-to-point circuit rented from an operator (e.g. an E1 or a fibre link). Guaranteed bandwidth, but expensive — the price grows with distance.
- **MPLS** (Multi-Protocol Label Switching): the operator's backbone forwards packets using short labels instead of full routing lookups, allowing guaranteed quality of service and private "virtual circuits" between a company's sites.
- **Satellite** (VSAT): covers rural areas with no terrestrial infrastructure (useful in the Far North

or East regions), but with high latency because the signal travels to a geostationary satellite about 35 786 km above the Earth and back.

- **4G/5G backhaul**: mobile operators such as MTN and Orange carry data from their base stations to their core network; organisations increasingly use 4G/5G routers as WAN access links, especially as backup.

PROPERTY **Characteristics of WAN links**

Compared with LAN links, WAN links typically have:

1. higher latency (a few ms inside a LAN; tens of ms across a country; 500–700 ms round trip via geostationary satellite);
2. lower bandwidth per unit cost (a gigabit LAN port is cheap; a gigabit inter-city circuit is very expensive);
3. the need for routing protocols (OSPF inside an organisation, BGP between operators) because data must cross many intermediate networks.

These properties are consequences of distance and shared infrastructure; no formal proof is examinable, but you must be able to justify them.

Criterion	LAN	MAN	WAN
Coverage	One building (≤ 1 km)	One city (5–50 km)	Country / continent
Typical owner	The organisation itself	Telecom operator	Operators (leased)
Latency	< 1 ms	1–10 ms	10–700 ms
Cost per Mbit/s	Very low	Medium	High
Key equipment	Switches	Metro switches, fibre rings	Routers

2.3 The Internet: a Global Public WAN

The largest WAN of all is the **Internet**: a network of networks interconnecting millions of LANs, MANs and WANs worldwide using the common TCP/IP protocol suite.

ISP hierarchy. The Internet is organised in tiers:

- **Tier 1** operators own global backbones and exchange traffic freely between themselves (peering).
- **Tier 2** operators (national or regional, e.g. CAMTEL, MTN Cameroon, Orange Cameroon) buy transit from Tier 1 and sell access to customers.
- **Tier 3** / access ISPs connect end users (homes, cybercafés, schools).

Two ISPs that exchange traffic without paying each other are said to peer; an ISP that pays a larger one to reach the whole Internet buys transit.

Routing with BGP. Each ISP is an **Autonomous System** (AS) identified by a number. The protocol **BGP** (Border Gateway Protocol) lets autonomous systems announce to each other which blocks of **public IP addresses** they can reach. When you load a website hosted abroad, BGP has already computed a path for your packets across several autonomous systems. Public IP addresses are allocated globally by IANA and regional registries (AFRINIC for Africa) so that every public address is unique worldwide.

PROPERTY **Metcalfe's Law**

The value of a telecommunications network is proportional to the square of the number n of its connected users:

$$V \propto n^2$$

Justification (instructive, not examinable as a proof): with n users, the number of possible one-to-one connections is the number of pairs $\binom{n}{2} = \frac{n(n-1)}{2} \approx \frac{n^2}{2}$, which grows like n^2. This explains why the Internet — and services like WhatsApp or mobile money — become dramatically more useful as more people join, and why connecting even a small Cameroonian school to the Internet plugs it into a network of billions of users.

WARNING **The Internet is a PUBLIC network**

A frequent exam mistake is to treat the Internet as a private WAN. It is public: packets cross routers owned by many different organisations, and anyone able to intercept them can read unencrypted data. Sensitive traffic (banking, ministry files, exam results) must therefore be protected with encryption: TLS for web applications (the padlock, HTTPS) and VPNs for site-to-site or remote communication.

2.4 Virtual Private Networks (VPN)

Intuition. A company in Yaoundé wants to use the cheap public Internet to reach its Douala branch, but without exposing its data. The solution is to build a private tunnel through the public network: packets are encrypted at one end and decrypted at the other, so interceptors see only unreadable ciphertext.

DEFINITION **Virtual Private Network**

A **VPN** is a technology that creates a secure, encrypted tunnel between two points over a public network (usually the Internet), giving the same confidentiality as a private leased line at a fraction of the cost. Two main types exist:

- Site-to-site VPN: two routers (or firewalls) encrypt all traffic between two whole networks, e.g. a head office and a branch. Users do nothing — the tunnel is permanent.

- Remote-access VPN: an individual user (a civil servant on mission, a teacher working from home) runs VPN software on a laptop or phone to connect securely to the organisation's network.

Tunnelling protocols.

- **IPsec**: the classic standard for site-to-site VPNs; works in two phases — Phase 1 (IKE) authenticates the two gateways and agrees on keys, Phase 2 negotiates the encryption parameters (e.g. AES) that protect the data itself.
- **SSL/TLS**: used by many remote-access VPNs; the same protocol that secures HTTPS, easy to deploy through firewalls.
- **WireGuard**: a modern, lightweight protocol with state-of-the-art cryptography and very fast performance.

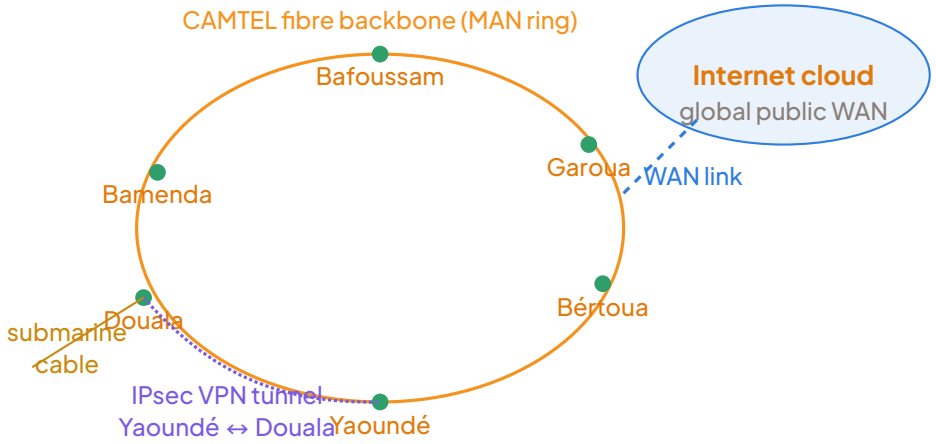
METHOD **Configuring a site-to-site IPsec VPN between two routers**

1. Plan: note the public IP address of each router (e.g. Yaoundé: 196.200.10.1, Douala: 196.200.20.1) and the private LAN behind each (e.g. 192.168.1.0/24 and 192.168.2.0/24).
2. Phase 1 (IKE): on both routers, define the peer's public IP, the same pre-shared key, and matching parameters (authentication method, encryption such as AES-256, hash such as SHA-256, Diffie-Hellman group). The two gateways authenticate each other and build a secure control channel.
3. Phase 2 (IPsec): define the traffic selectors (which subnets may use the tunnel: 192.168.1.0/24 ↔ 192.168.2.0/24) and the encryption/integrity algorithms for the data channel.
4. Activate and test: bring the tunnel up, then ping a machine in the remote LAN and check with the router's status command that Phase 1 and Phase 2 security associations are established.
5. Verify security: confirm that traffic between the sites is encrypted (a packet capture on the WAN interface must show only ESP ciphertext, never readable data).

DID YOU KNOW? **Cameroon: government and enterprise VPNs**

In Cameroon, ministries and public enterprises interconnect their services through secure links over CAMTEL's infrastructure, and many banks and companies (including MTN and Orange for their internal systems) use site-to-site IPsec VPNs between their Yaoundé headquarters and Douala branches. ANTIC, the National Agency for Information and Commu-

nication Technologies, promotes exactly these good practices — encryption of sensitive exchanges — in its national cybersecurity mission.



Simplified map of Cameroon: a metropolitan fibre ring interconnects the major cities (MAN), a WAN link joins the Internet cloud, and an encrypted IPsec VPN tunnel protects traffic between the Yaoundé and Douala offices.

WORKED EXAMPLE 2 **Choosing between a leased line and a VPN**

Statement. A microfinance institution has its head office in Yaoundé (LAN 192.168.10.0/24) and a branch in Douala (LAN 192.168.20.0/24). Option A: lease a dedicated 10 Mbit/s line between the two cities. Option B: buy ordinary Internet access at both sites and build a site-to-site IPsec VPN. Compare the two options and recommend one.

SOLUTION

Solution.

- Option A (leased line): guaranteed bandwidth and low, stable latency; traffic never crosses the public Internet, so it is naturally private. However, a dedicated inter-city circuit is very expensive, and the cost grows with distance.
- Option B (VPN over Internet): uses cheap Internet subscriptions; IPsec Phase 1 authenticates the two routers with a pre-shared key and Phase 2 encrypts all traffic between 192.168.10.0/24 and 192.168.20.0/24 with AES, so confidentiality is preserved even though packets cross the public network. Latency and bandwidth depend on the Internet and are less predictable.

Recommendation: for a small institution with a limited budget, Option B offers the best cost/security compromise, provided a strong pre-shared key and AES encryption are

used. A bank's core transaction system, which needs guaranteed quality, might justify Option A or an MPLS service.

EXERCISE 1 **Latency and Metcalfe's Law**

1. A geostationary satellite is about 35 786 km above the Earth. A signal travels at about 3×10^8 m/s. Estimate the minimum round-trip time (Earth → satellite → Earth, twice for a request and its reply) and explain why VSAT feels "slow" for web browsing.
2. A messaging service has $n = 1000$ users in a town. Using Metcalfe's Law, by what factor does the value of the network increase if the number of users doubles? Comment for a mobile-money operator in Cameroon.

SOLUTION **Exercise 1**

1. One-way path Earth → satellite: 35 786 km $\approx 3.58 \times 10^7$ m, taking $\frac{3.58 \times 10^7}{3 \times 10^8} \approx 0.12$ s. A full request-and-reply cycle crosses this path four times: $4 \times 0.12 \approx 0.48$ s, i.e. about 500 ms before any processing delay. Every click therefore waits at least half a second, which is why satellite links feel slow despite decent bandwidth.
2. By Metcalfe's Law $V \propto n^2$, so doubling n multiplies V by $2^2 = 4$: the network's value quadruples while the user base only doubles. This is why mobile-money operators fight to grow their subscriber base: each new customer makes the service more useful for all existing customers.

KEY POINTS **Key points**

- A **MAN** covers a city (5–50 km) and is usually run by a telecom operator (Metro Ethernet, fibre ring, WiMAX).
- A **WAN** covers countries and continents by leasing operator services (leased lines, MPLS, satellite, 4G/5G backhaul); it has higher latency and higher cost per bandwidth than a LAN and requires routing protocols.
- The **Internet** is a global public WAN: ISPs organised in tiers, interconnected by peering and transit, routed by BGP, using unique public IP addresses.
- Because the Internet is public, sensitive traffic must be encrypted: **VPN** tunnels (site-to-site or remote-access) using IPsec, SSL/TLS or WireGuard.

- Metcalfe's Law: network value grows like n^2 — the reason large networks such as the Internet and mobile money become indispensable.

3 CHOOSING THE RIGHT NETWORK FOR AN ORGANISATION

Having examined Metropolitan, Wide Area Networks and the Internet, we now turn to the practical task of selecting the most appropriate network architecture for a given organisation. The choice is never arbitrary; it follows from a systematic analysis of requirements, constraints and future growth.

3.1 Key Decision Criteria

Before comparing network types, the decision-maker must quantify the following parameters for the organisation:

- **Geographic spread**: distance between sites (same building, same city, different regions, different countries).
- **Number of users**: total concurrent devices (workstations, printers, IoT sensors, mobile phones).
- **Bandwidth needs**: peak and average data rates per user and per application (VoIP, video conferencing, cloud backup, ERP).
- **Security requirements**: regulatory compliance (e.g. Cameroon's data protection legislation and ANTIC guidelines), sensitivity of data, need for encryption and intrusion detection.
- **Budget**: capital expenditure (CAPEX) for cabling, switches, routers, firewalls; operational expenditure (OPEX) for leased lines, ISP subscriptions, maintenance contracts.
- **Scalability**: ease of adding sites, users or bandwidth without a complete redesign.
- **Mobility**: proportion of remote or roaming users requiring secure access from home, client sites or public hotspots.

Each criterion can be weighted according to organisational priority. A weighted decision matrix (see the Method box below) turns these qualitative factors into a quantitative score for each candidate network type.

3.2 Mapping Criteria to Network Types

The table below summarises the typical fit between organisational profiles and network categories. It is a guideline, not a rigid rule; hybrid solutions are common.

Organisational Profile	Recommended Network Type	Typical Technologies	Key Advantage
Single office, <50 users, high local traffic	**LAN**	Twisted-pair (Cat6/6a), Gigabit/10GbE switches, Wi-Fi 6	Low latency, full control, inexpensive per port
Multiple buildings on a campus (university, hospital)	**CAN**	Fibre backbone (single-mode), distribution switches, VLAN segmentation	High bandwidth between buildings, centralised management
Several sites within a metropolitan area (e.g. Yaoundé branches)	**MAN**	Metro-Ethernet, dark fibre lease, MPLS from CAMTEL/MTN	Guaranteed SLAs, symmetric bandwidth, carrier-grade reliability
National or international sites, need for secure inter-site traffic	**WAN / Site-to-Site VPN**	IPsec over broadband/4G/5G, MPLS, SD-WAN overlay	Cost-effective, encrypted, flexible transport
Remote / mobile workers, occasional access to internal resources	**Remote-Access VPN**	SSL/TLS VPN, IKEv2, Zero-Trust Network Ac…	Strong authentication, granular…

3.3 Cost Analysis: CAPEX vs OPEX

A realistic budget compares one-time capital outlay with recurring operational costs. The following simplified model illustrates the trade-off for a three-site organisation (Yaoundé, Douala, Bafoussam) choosing between a dedicated MAN (leased fibre) and a VPN over commercial broadband with 4G backup. Figures are indicative estimates for the Cameroonian market.

Cost Element	Dedicated MAN (Leased Fibre)	VPN over Broadband + 4G Backup
CAPEX (Year 0)	Fibre termination equipment, routers, managed switch stack (~FCFA 12 000 000)	VPN concentrators, firewalls, 4G routers (~FCFA 4 500 000)
OPEX (Annual)	Leased line fees (3 sites × FCFA 3 500 000) + maintenance (FCFA 1 200 000) = FCFA 11 700 000	ISP broadband (3 × FCFA 1 200 000) + 4G backup (3 × FCFA 600 000) + VPN licence (FCFA 800 000) = FCFA 5 600 000
5-Year Total	FCFA 12 000 000 + 5 × 11 700 000 = FCFA 70 500 000	FCFA 4 500 000 + 5 × 5 600 000 = FCFA 32 500 000

The VPN option is roughly half the 5-year cost, but the MAN offers deterministic latency and a contractual SLA. The final decision hinges on how critical guaranteed performance is for the organisation's core applications.

PROPERTY **Leased Lines vs Broadband**

A **leased line** (dedicated fibre, Metro-Ethernet, MPLS) provides guaranteed bandwidth, symmetric upload/download, and a Service Level Agreement (SLA) with penalties for downtime. **Broadband** (ADSL, VDSL, 4G/5G, cable) is a best-effort service: bandwidth is shared, asymmetric, and no formal SLA is usually offered. For mission-critical real-time applications (voice, video, SCADA), leased lines are preferred; for general office traffic, broadband with a VPN overlay is often sufficient.

3.4 Redundancy and Reliability

A single link or device constitutes a **single point of failure (SPOF)**. Reliability engineering introduces redundancy at multiple layers:

- **Link redundancy**: two independent ISP connections (e.g. fibre + 4G) per site, terminated on separate routers.
- **Device redundancy**: dual firewalls / routers in active-standby (VRRP/HSRP) or active-active (load-sharing) mode.
- **Path redundancy**: dynamic routing protocols (OSPF, BGP) or SD-WAN policies that detect link degradation and failover within seconds.
- **Power redundancy**: UPS and generator for network cabinets.

A well-designed failover test plan (periodic simulated link failure) validates that the recovery time objective (RTO) meets business requirements.

3.5 Future-Proofing: IPv6, Cloud Integration, SD-WAN

Modern networks must accommodate:

- **IPv6 readiness**: dual-stack deployment on all edge devices; ISP support for IPv6 prefix delegation.
- **Cloud integration**: direct peering (e.g. AWS Direct Connect, Azure ExpressRoute) or secure Internet breakout at each branch (local Internet breakout) to avoid hair-pinning traffic through headquarters.
- **SD-WAN**: a software-defined overlay that abstracts underlying transports (MPLS, broadband, 4G/5G) and applies application-aware policies for dynamic path selection, QoS, and zero-touch provisioning.

DEFINITION **SD-WAN**

SD-WAN (Software-Defined Wide Area Network) is an architecture that separates the control plane from the data plane, allowing a centralised orchestrator to manage multiple transport links (MPLS, broadband, LTE) as a single logical WAN. It provides application-aware routing, automatic failover, encryption, and zero-touch provisioning of branch appliances.

DID YOU KNOW? **ANTIC and Network Security in Cameroon**

The **Agence Nationale des Technologies de l'Information et de la Communication (ANTIC)** is the regulatory body overseeing cybersecurity and electronic certification in Cameroon. Organisations designing networks must comply with ANTIC guidelines on data protection, encryption standards, and incident reporting. The Law No. 2010/012 on cybersecurity and cybercrime, along with the recent personal data protection law, impose obligations for securing network infrastructure, especially when handling citizens' personal data. Financial institutions and telecom operators face additional sector-specific regulations from COBAC and ART respectively.

WARNING **Over-Provisioning a MAN**

A common mistake is to lease a Metro-Ethernet MAN for an organisation with only two sites in the same city. The monthly recurring cost is often 3–5× higher than a site-to-site IPsec VPN over business-grade broadband with a 4G backup. Unless the SLA, symmetric bandwidth, or regulatory mandate strictly requires a MAN, the VPN solution is far more cost-effective.

EXEMPLE **Cameroonian NGO Network Design**

Scenario: GreenFuture operates offices in Yaoundé (HQ, 80 users), Douala (45 users) and Bafoussam (20 users). They run a cloud-based ERP, VoIP, and need secure access for 15 field agents.
Recommended solution:

1. LAN at each site: Cat6a cabling, PoE+ switches, Wi-Fi 6 APs, VLANs for voice, data, guest.
2. Site-to-site IPsec VPN over business broadband (CAMTEL/MTN) with 4G LTE backup at each site. VPN concentrators at HQ and each branch.
3. Remote-access SSL VPN with MFA for field agents.
4. SD-WAN overlay (e.g. FortiGate SD-WAN or Cisco Meraki) to steer ERP traffic over the best path, enforce QoS for VoIP, and provide zero-touch deployment for future sites.
5. IPv6 dual-stack on all edge routers; cloud breakout at each branch for Microsoft 365 / Google Workspace.

Estimated 3-year TCO: FCFA 28 000 000 (vs. FCFA 62 000 000 for a full MAN).

METHOD **Decision Matrix Template**

Step 1: List criteria (geographic spread, users, bandwidth, security, budget, scalability, mobility). Assign a weight (sum = 100).
Step 2: For each candidate network type (LAN, CAN, MAN, WAN/VPN, Remote-Access VPN), score 1–5 on each criterion.
Step 3: Multiply score × weight, sum across criteria to obtain a weighted total.
Step 4: Select the network type with the highest total. If two are close, analyse qualitative factors (vendor lock-in, staff expertise, regulatory).
Step 5: Document the rationale; revisit annually or when major changes occur.

WORKED EXAMPLE 3 **Worked Example: Selecting a Network for a Growing Startup**

Statement: TechHub is a Cameroonian fintech startup. Currently 30 developers work in a single open-space office in Douala. They plan to open a second office in Yaoundé (15 developers) within 6 months, and hire 10 remote contractors across the country. Their application requires low-latency access to a PostgreSQL database hosted on a cloud VM (Azure West Europe). Budget is tight (FCFA 8 000 000 CAPEX, FCFA 3 000 000/year OPEX). Choose the most suitable network architecture and justify.

SOLUTION

Solution:

1. Identify criteria and weights: Geographic spread (25), Users (10), Bandwidth (20), Security (20), Budget (15), Scalability (5), Mobility (5). Sum = 100.
2. Candidate types: (A) Single LAN + remote-access VPN; (B) Two LANs + site-to-site IPsec VPN over broadband; (C) MAN (Metro-Ethernet) between Douala and Yaoundé.
3. Score each (1–5):

Criterion	A	B	C
Geographic spread	2	5	4
Users	4	5	5
Bandwidth	3	4	5
Security	3	5	5
Budget	5	4	1
Scalability	3	5	3
Mobility	4	4	2

4. Weighted totals:

   A = 2×25 + 4×10 + 3×20 + 3×20 + 5×15 + 3×5 + 4×5 = 50 + 40 + 60 + 60 + 75 + 15 + 20 = 320

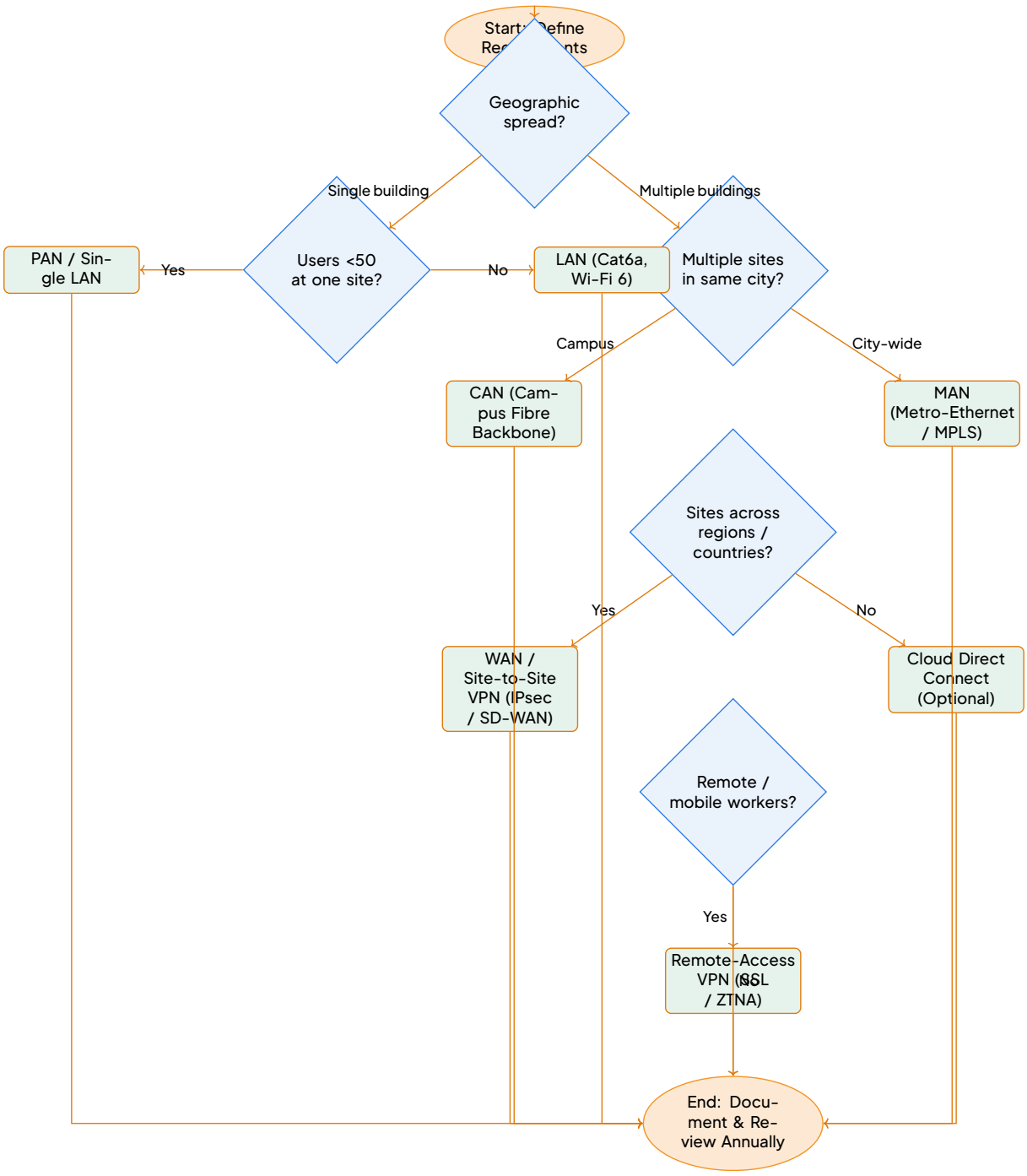
   B = 5×25 + 5×10 + 4×20 + 5×20 + 4×15 + 5×5 + 4×5 = 125 + 50 + 80 + 100 + 60 + 25 + 20 = 460

   C = 4×25 + 5×10 + 5×20 + 5×20 + 1×15 + 3×5 + 2×5 = 100 + 50 + 100 + 100 + 15 + 15 + 10 = 390

5. Decision: Option B (two LANs + site-to-site IPsec VPN) scores highest (460). It fits the budget, provides encryption for financial data, scales to future sites, and supports remote contractors via the same VPN concentrators.
6. Implementation sketch: Deploy Cat6a LANs at both offices, FortiGate 60F at each site for IPsec VPN (AES-256, SHA-256), 4G failover, Azure VPN Gateway for cloud connectivity, SSL VPN with MFA for contractors. Estimated CAPEX: FCFA 6 500 000; OPEX: FCFA 2 800 000/year.

3.6 Decision-Tree Flowchart for Network Selection

The following flowchart guides the network architect from the simplest to the most complex choice based on the key organisational parameters discussed above.



Decision-tree flowchart for selecting the appropriate network type based on organisational parameters.

3.7 Summary of Key Points

KEY POINTS **Choosing the Right Network**

- Map every requirement (geography, users, bandwidth, security, budget, scalability, mobility) to a weighted decision matrix.
- Small, single-site organisations → LAN (or PAN for personal devices).
- Campus-style multi-building → CAN with fibre backbone.
- City-wide multi-site with SLA needs → MAN (leased fibre / Metro-Ethernet).
- National / international sites → WAN built from site-to-site VPN over broadband/4G/5G, optionally orchestrated by SD-WAN.
- Remote workers → Remote-access VPN (SSL/TLS, IKEv2, ZTNA) with MFA.
- Always compare CAPEX vs OPEX over a 3–5 year horizon; include redundancy (dual ISP, 4G backup, VRRP).
- Future-proof with IPv6 dual-stack, cloud direct connect / local Internet breakout, and SD-WAN for dynamic path selection.
- Avoid over-provisioning: a two-site city deployment rarely justifies a full MAN; a VPN overlay is usually cheaper and sufficiently reliable.

The next section (Integration Activity – Network Design Project) will give you the opportunity to apply these principles in a complete, graded case study.

4 INTEGRATION ACTIVITY – NETWORK DESIGN PROJECT

4.1 From Theory to Practice: A Real-World Design Challenge

In the previous sections, we learned how to classify networks by coverage area (Lesson 10) and how to select the appropriate network type for an organisation by analysing coverage, budget, performance, and security requirements (Lesson 11). Now, we put that knowledge into action. This integration activity (Lesson 12) simulates a professional ICT consultancy task: designing a network infrastructure for a regional healthcare system in the Northwest Region of Cameroon. You will work in teams to produce a complete design dossier, defend your choices, and evaluate peers' work using a structured rubric. This mirrors the kind of project you may face in higher education or as a junior network engineer at organisations like CAMTEL, MTN, Orange, or the Ministry of Public Health.

4.2 Case Study: Northwest Regional Hospital Network (NRHN)

EXPERIMENT **Case Study Brief**

The Ministry of Public Health wishes to modernise the information system of the North-west Regional Health Delegation. The network must connect five sites:

- Site A: Regional Hospital Bamenda (administrative headquarters, 300+ staff, high-density Wi-Fi, server room).
- Site B: District Hospital Kumbo (150 staff, radiology department, needs DICOM image transfer).
- Site C: District Hospital Ndop (120 staff, tele-medicine pilot with specialists in Bamenda).
- Site D: District Hospital Wum (80 staff, remote, limited fibre availability).
- Site E: Batibo Integrated Health Centre (30 staff, basic admin and patient records).

Functional requirements:

1. Centralised Electronic Medical Record (EMR) database hosted at Site A, accessed in real time from all sites.
2. Tele-medicine video conferencing between Site A (specialists) and Sites B, C, D (consultation rooms).
3. Administrative voice over IP (VoIP) and file sharing across all sites.
4. Internet access for all sites (e-mail, research, cloud backup).
5. Segregation of traffic: medical devices (IoT), clinical workstations, administrative PCs, guest Wi-Fi.

Constraints:

- Budget: 25 million FCFA capital expenditure (CAPEX), 3 million FCFA/year operational expenditure (OPEX).
- Sites B–E are 40–80 km from Bamenda; terrain is hilly, some areas lack reliable fibre.
- Power supply is unstable; Sites D and E experience daily outages of 4–6 hours.
- Compliance with Cameroon's Law No. 2010/012 on cybersecurity and cybercrime and ANTIC guidelines.

4.3 Organising the Team: Roles and Responsibilities

METHOD **Structured Group Work Roles**

Team size: 4 students. Each member takes a primary role but contributes to all deliver-

ables.

1. Requirements Analyst: Interviews stakeholders (teacher acts as client), writes the Requirements Specification Document (RSD), lists functional/non-functional requirements, prioritises them (MoSCoW method).
2. Network Designer: Selects network type per site (LAN, CAN) and inter-site links (MAN, WAN, VPN), draws the logical diagram, justifies each choice with syllabus criteria (coverage, bandwidth, latency, cost).
3. Cost Estimator: Researches current Cameroonian market prices (CAMTEL, MTN Business, Orange Business, local vendors) for cabling, switches, routers, wireless APs, fibre/microwave links, VPN appliances, UPS/generators. Produces a cost estimate table (CAPEX/OPEX) within budget.
4. Security & Presentation Lead: Defines security measures (firewall rules, VPN tunnels, VLAN segmentation, access control), prepares the final slide deck (10 minutes), coordinates peer review responses.

Deliverables (due in 3 weeks):

- RSD (2–3 pages).
- Logical network diagram (A3, digital).
- Cost estimate spreadsheet (printed + digital).
- Security policy summary (1 page).
- Presentation slides (PDF).

4.4 Deliverable 1: Requirements Specification

The analyst leads this, but the whole team validates. Use the template below. A good RSD distinguishes **Must have**, **Should have**, **Could have**, **Won't have** (MoSCoW).

EXEMPLE **Excerpt from a Requirements Specification**

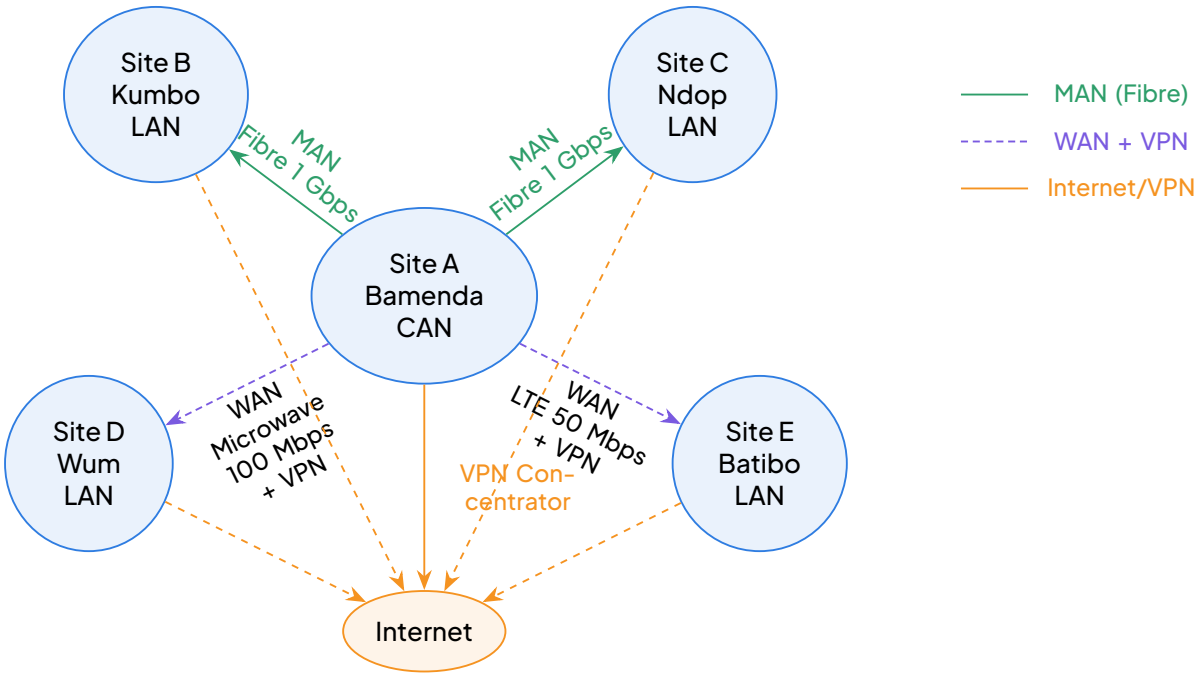
ID	Requirement	Priority
FR-01	Real-time access to central EMR database from all clinical workstations	Must
FR-02	Video conferencing (1080p, 30 fps) for tele-medicine between Site A and Sites B, C, D	Must
FR-03	VoIP for administrative calls across all sites	Should
NFR-01	End-to-end latency < 100 ms for EMR transactions	Must
NFR-02	Bandwidth ≥ 50 Mbps per site for clinical traffic	Must
NFR-03	Network availability ≥ 99.5% (excluding power outages)	Should
NFR-04	Compliance with ANTIC cybersecurity framework	Must

4.5 Deliverable 2: Network Type Selection and Logical Diagram

The designer maps each site to a network class and chooses inter-site technology. Justification must reference coverage area definitions (Lesson 10) and selection criteria (Lesson 11): bandwidth, latency, reliability, cost, scalability, security.

PROPERTY **Network Classification for NRHN Sites**

- Site A (Bamenda): **Campus Area Network (CAN)** – multiple buildings (wards, admin, server room) within 1 km, high-speed fibre backbone, managed switches, wireless controllers.
- Sites B, C, D, E: **Local Area Network (LAN)** each – single building or small compound, structured cabling (Cat6a), PoE switches for APs and VoIP phones.
- Inter-site links (A↔B, A↔C, A↔D, A↔E): **Metropolitan Area Network (MAN)** where fibre exists (Bamenda–Kumbo, Bamenda–Ndop) – leased dark fibre or CAMTEL Metro Ethernet. For Wum and Batibo (no fibre): **Wide Area Network (WAN)** via microwave radio links or 4G/5G LTE backup with **Site-to-Site VPN** over public Internet for encryption.
- Internet access: Each site has a local ISP link (fibre or LTE) with a **VPN concentrator** at Site A for centralised security inspection.



Logical Network Diagram Template for NRHN. Sites are LAN/CAN clouds; inter-site links labelled with technology and VPN where public infrastructure is used.

4.6 Deliverable 3: Cost Estimate Table

The cost estimator researches real prices (2024–2025) from Cameroonian vendors. Prices are indicative; students must cite sources (vendor quotes, websites).

WORKED EXAMPLE 4 **Worked Example: Cost Estimate for Inter-Site Links (CAPEX)**

Statement: Estimate the capital cost for the inter-site connectivity of NRHN, assuming:

- Bamenda–Kumbo (45 km): CAMTEL dark fibre lease, 1 Gbps, 2 km trenching at Site B.
- Bamenda–Ndop (40 km): CAMTEL Metro Ethernet, 1 Gbps, no trenching.
- Bamenda–Wum (75 km): Microwave link (licensed band), 100 Mbps, towers at both ends.
- Bamenda–Batibo (60 km): LTE router + outdoor antenna, 50 Mbps, VPN appliance at Site A.

Solution:

Link	Components	Qty	Unit Cost (FCFA)	Total (FCFA)
Bamenda–Kumbo	Dark fibre lease (5 yr IRU), 1 Gbps	1	4,500,000	4,500,000
	Trenching & ducting (2 km)	1	1,200,000	1,200,000
	Fibre patch panels, SFP+ modules	2	85,000	170,000
Bamenda–Ndop	Metro Ethernet 1 Gbps (5 yr)	1	3,800,000	3,800,000
	SFP+ modules	2	85,000	170,000
Bamenda–Wum	Microwave radios (licensed, 100 Mbps)	2	1,500,000	3,000,000
	Towers (15 m) & installation	2	600,000	1,200,000
	Antennas, cables, surge protectors	2	150,000	300,000
Bamenda–Batibo	Industrial LTE router + MIMO antenna	1	350,000	350,000
	VPN appliance (Site A, 5 tunnels)	1	1,200,000	1,200,000
	Configuration & commissioning	1	200,000	200,000
Subtotal Inter-Site CAPEX				14,890,000

Comment: This leaves 10 million FCFA for LAN/CAN equipment, servers, UPS, licensing, and contingency. OPEX includes annual fibre lease renewals, microwave licence fees (ART), LTE data plans (150,000 FCFA/month/site), and maintenance contracts.

SOLUTION

Detailed Solution Steps:

1. Identify each link's technology based on coverage and availability (Lesson 10, 11).
2. Obtain at least two quotes per major item; use the average.
3. Include ancillary costs: installation, configuration, licensing, surge protection (critical in Cameroon).

4. Sum and verify against budget; if over, iterate (e.g., reduce Wum link to 50 Mbps LTE only, save 2M FCFA).
5. Document assumptions: exchange rate, warranty period, vendor support SLA.

4.7 Deliverable 4: Security Measures – Firewall, VPN, VLAN

DEFINITION **VLAN (Virtual Local Area Network)**

A VLAN is a logical segmentation of a physical switch into multiple broadcast domains. Devices in different VLANs cannot communicate directly at Layer 2; they require a router or Layer 3 switch. In NRHN, we define at least four VLANs per site:

- VLAN 10 – Medical Devices: Patient monitors, imaging modalities (DICOM), infusion pumps. No Internet access, strict ACLs.
- VLAN 20 – Clinical Workstations: Doctors/nurses PCs accessing EMR. Access to VLAN 10 (read-only for some), Internet via proxy.
- VLAN 30 – Administration: HR, finance, VoIP phones. Access to file servers, Internet, no access to VLAN 10.
- VLAN 40 – Guest Wi-Fi: Patients/visitors. Internet only, rate-limited, isolated via captive portal.

Inter-site VPN tunnels terminate on a central firewall at Site A, which enforces inter-VLAN routing policies.

KEY POINTS **Security Checklist for NRHN**

- Perimeter: Next-generation firewall at Site A (IPS, anti-malware, SSL inspection). Branch firewalls at Sites B–E (managed centrally).
- VPN: IPsec site-to-site tunnels (AES-256, SHA-256, DH Group 14) for all inter-site traffic. Remote access VPN (MFA) for doctors working from home.
- Segmentation: VLANs as above; trunk links between switches carry tagged VLANs (802.1Q). Access ports assigned via 802.1X or MAC authentication bypass.
- Monitoring: Syslog/SIEM (e.g., Wazuh) collecting logs from all firewalls, switches, servers. Alert on failed logins, port scans, unusual traffic.
- Backup: Encrypted off-site backup of EMR database (daily incremental, weekly full) to a cloud provider in Douala or Yaoundé.

- Compliance: Data residency (patient data stays in Cameroon), ANTIC incident reporting procedures, annual penetration test.

4.8 Peer Review and Rubric

Each team evaluates another team's dossier using the rubric below. Scores are discussed in a plenary session moderated by the teacher.

Criterion	Description	Weight
Coverage Justification	Each site and link correctly classified (LAN, CAN, MAN, WAN, VPN) with syllabus-based reasoning (distance, bandwidth, ownership, geography).	30%
Diagram Clarity	Logical diagram uses standard symbols, shows all sites, links, VLANs, VPN tunnels, Internet edge. Labels are readable; legend included.	25%
Cost Realism	Prices sourced from Cameroonian vendors (quotes/websites cited). CAPEX ≤ 25M FCFA, OPEX ≤ 3M FCFA/yr. Contingency (10%) shown.	20%
Security Completeness	Firewall, VPN, VLAN segmentation, access control, monitoring, backup, compliance addressed. No major gap (e.g., missing guest isolation).	15%
Presentation	Clear structure, confident delivery, answers questions well, all members participate. Slides professional (consistent theme, readable fonts).	10%

4.9 Teacher Debrief: Linking Design Choices to Syllabus Concepts

After presentations, the teacher highlights how each decision maps to the chapter's lessons:

- **Lesson 10 (Types by Coverage Area):** The classification of Site A as CAN (multiple buildings, same organisation, < 5 km) vs Sites B–E as LAN (single building). The inter-site links as MAN (fibre within city/metro area) and WAN (microwave/LTE over longer distances, public infrastructure).
- **Lesson 11 (Choosing the Right Network):** The selection criteria table (bandwidth, latency, cost, reliability, security, scalability) was applied to each link. Example: Fibre chosen for Kumbo/Ndop because latency < 5 ms required for DICOM; microwave for Wum because fibre not available, but licensed band ensures reliability; LTE+VPN for Batibo as cost-effective fallback.
- **VPN as a WAN extension:** VPN tunnels over public Internet (LTE) effectively create a virtual MAN/WAN, encrypting traffic – a key concept for modern hybrid networks.
- **Lesson 12 (Integration Activities):** This project synthesises classification, selection, costing, security, and presentation skills into a single coherent deliverable.

- **Integration of security:** VLANs (Layer 2 segmentation) and firewalls (Layer 3/4/7) illustrate defence in depth, linking to the OSI model (see Property box).

4.10 Critical Practical Considerations

WARNING **Power Backup – The Silent Network Killer**

In Cameroon, especially in remote areas like Wum and Batibo, network availability is meaningless without electricity. A common student mistake is to design a perfect network but forget:

- UPS (Uninterruptible Power Supply): At least 2 hours autonomy for core switches, routers, firewall, server rack at each site. Online double-conversion UPS for Site A server room.
- Generator: For Sites D and E, a 10–15 kVA diesel/solar hybrid generator with automatic transfer switch (ATS) to cover 4–6 hour daily outages.
- PoE Budget: Ensure switches have enough PoE+ budget for APs, VoIP phones, IP cameras on UPS-backed ports.
- Monitoring: SNMP traps for UPS battery low, generator start/stop, mains failure.

Exam tip: If a question asks "What factors affect network availability in a rural Cameroonian context?", power backup is a top answer.

PROPERTY **OSI Model Layers in Network Design Decisions**

Every design choice operates at one or more OSI layers. Examiners may ask you to identify the layer.

Layer	NRHN Design Decision	Key Protocols / Technologies
1 Physical	Fibre type (OS2), microwave radio frequency, LTE bands, UPS/generator	Ethernet (1000BASE-LX), RF, 4G/5G NR
2 Data Link	VLAN tagging (802.1Q), trunk/access ports, MACsec for link encryption	Switching, STP/RSTP, LLDP
3 Network	IP addressing plan (VLSM), routing (OSPF/BGP), VPN tunneling (IPsec)	IPv4/IPv6, OSPF, BGP, IPsec ESP
4 Transport	TCP for EMR/VoIP signalling, UDP for video streaming (RTP), QoS marking (DSCP)	TCP, UDP, SCTP, DSCP EF/AF41
5 Session	SIP for VoIP call control, TLS session resumption for VPN	SIP, TLS 1.3
6 Presentation	DICOM encoding, HL7/FHIR for EMR, video codecs (H.264/H.265)	DICOM, HL7, FHIR, H.265
7 Application	EMR web interface, telemedicine app, VoIP softphone, backup agent	HTTPS, proprietary apps

Examinable: You must be able to map a requirement (e.g., "segment medical devices from admin traffic") to the correct layer (Layer 2 – VLAN) and technology (802.1Q).

4.11 Final Reflection

This integration activity is not just an exam exercise; it is a portfolio piece. Keep your dossier (RSD, diagram, cost sheet, security policy, presentation) – it demonstrates to universities and employers that you can:

- Translate vague organisational needs into technical specifications.
- Apply the network classification framework (LAN, CAN, MAN, WAN, VPN) to a real Cameroonian geography.
- Balance performance, cost, and security within a constrained budget.
- Communicate technical decisions clearly to non-technical stakeholders.

The skills you practised here – requirements analysis, logical design, costing, security hardening, peer review – are the foundation of the **Network Administrator** and **Systems Engineer** roles in Cameroon's growing digital economy (Silicon Mountain, e-government projects, fintech). Approach the GCE Advanced Level practical paper with the same structured mindset, and you will excel.

DID YOU KNOW? **Did You Know?**

The National Agency for Information and Communication Technologies (ANTIC) in Cameroon runs the Cybersecurity Awareness Campaign and operates the CERT-CM (Computer Emergency Response Team). Any health information system like NRHN must register with ANTIC and follow the National Cybersecurity Strategy. In 2023, ANTIC reported a rise in ransomware targeting healthcare facilities – hence the emphasis on offline backups and network segmentation in this project.

5 INTEGRATION ACTIVITY

6 INTEGRATION ACTIVITY: NETWORK DESIGN PROJECT

6.1 Aims of the activity

This integration activity closes the chapter on the **classification of computer networks**. It places you in the role of an ICT consultant and asks you to mobilise — not simply recall — everything studied in the chapter:

- classify networks by **coverage area** (PAN, LAN, CAN, MAN, WAN) and justify each classification;
- apply the **selection criteria** studied in Lesson 11 (size and distance, number of users, required services and bandwidth, budget, reliability, security, scalability);
- choose appropriate **transmission media** and **interconnection technologies** (fibre optic, leased line, VPN over the public Internet or 4G);
- design and draw a **logical network diagram** for a multi-site organisation;
- estimate a realistic **budget** and propose **security controls** adapted to sensitive data (here, medical records);
- defend technical choices in writing and orally, and evaluate the work of peers.

KEY POINTS **Competence targeted**

By the end of this activity you should be able to take any real organisation (school, bank, hospital, ministry), analyse its needs, and produce a justified network design in which every choice — network type, medium, topology, security — is argued from the constraints of the situation. This is exactly the kind of reasoning expected in the GCE Advanced Level paper.

6.2 Situation

Read the following brief carefully. All the data you need is contained in it.

EXPERIMENT **Client brief — Northwest Regional Health Network (NWRHN)**

The Regional Delegation of Public Health for the Northwest Region wishes to interconnect five health facilities so that they can share an **Electronic Health Record** (EHR) system, hold **video consultations** (telemedicine) with specialists, and use **VoIP** telephony

between sites to reduce telephone bills. You are the ICT consultant engaged to propose the network design.
The five sites

Site	Des	Staff / PCs	Distance from Bamenda
A — Regional Hospital (Bamenda)	Main re-fer-ral hos-pi-tal; hosts the EHR server and the fu-ture data room; two build-ings 300 m apart on one cam-pus	180 users, 120 PCs	0 km
B — District Hospital (Ndop)	Distr hos-pi-tal, one build-ing	45 users, 25 PCs	40 km
C — District Hospital (Kumbo)	Distr hos-pi-tal, one build-ing	50 users, 30 PCs	110 km
D — District Hospital (Wum)	Distr hos-pi-tal, one build-ing	30 users, 15 PCs	80 km

Available connectivity options (as offered locally)

- CAMTEL fibre is available in Bamenda and along the Bamenda–Kumbo road (sites A, B, C): metropolitan fibre or leased lines, 10 to 100 Mbit/s, monthly subscription, high cost.
- MTN / Orange 4G covers all five towns, with business data bundles; quality is good in Bamenda and Kumbo, average in Ndop and Wum, weak but usable in Nkambe.
- VSAT satellite is possible anywhere but expensive and has high latency.

Required services and constraints

- The EHR database (hosted at site A) must be reachable from all sites at any time; medical records are confidential by nature.
- Video consultation between any district hospital and Bamenda requires about 2 Mbit/s per session with low delay.
- VoIP between sites requires little bandwidth but is sensitive to delay and packet loss.
- The Delegation's budget is limited: the solution must be proportionate — the best technical solution is not acceptable if it is unaffordable.
- The network must be able to grow (two more health centres may join within five years).

The activity runs over **two sessions of 90 minutes**. Session 1: analysis and design in groups of three or four. Session 2: peer review — each group presents its diagram and budget in 5 minutes; the other groups challenge the choices using the selection criteria of Lesson 11; each group then amends its design.

6.3 Tasks

1. **Classify.** For each site, state the type of network needed inside the site (PAN, LAN or CAN), justifying with the size of the site and the number of users. Pay special attention to site A, which has two buildings.
2. **Interconnect.** The five sites together form which class of network? Justify your answer using the notion of coverage area.
3. **Choose media and technologies.** For each link between a site and the rest of the network, choose a technology among: CAMTEL fibre/leased line, VPN over 4G, VSAT. Justify each choice with at least two criteria (bandwidth, reliability, cost, availability at that location).
4. **Explain the VPN.** For the sites connected through the public 4G network, explain in four or five lines what a VPN is, why it is indispensable here, and what it protects (think of the confidentiality of medical records).
5. **Draw.** Produce a logical diagram of the whole network showing: the five sites, the internal network of site A (two buildings, server room, switches), the interconnection links with their technologies, and the security devices (firewalls). Use standard symbols.

6. **Budget.** Complete a cost table with realistic estimated figures (in FCFA): equipment per site (router, switch, firewall), cabling at site A, and monthly subscriptions. Distinguish investment (once) from recurrent costs (monthly).
7. **Secure.** Propose three security controls adapted to this hospital network and say which threat each one addresses.
8. **Plan for growth.** Explain how your design accommodates two future health centres without being rebuilt.
9. **Peer review.** During session 2, note two strengths and one weakness of another group's design, using the vocabulary of the chapter (coverage area, bandwidth, reliability, scalability, security).

WARNING **Common traps**

- Do not call the inter-site network a "LAN": sites 40 to 130 km apart can never form a LAN.
- Do not propose fibre everywhere "because it is the best": a design that ignores the budget and the actual availability of fibre at Wum or Nkambe is not a valid engineering answer.
- A VPN is not a network type by coverage area; it is a security technique that creates a private tunnel over a public network.
- Distinguish clearly the physical level (cables, 4G, fibre) from the logical level (who talks to whom, through which tunnel).

6.4 Model answer

1. Classification inside each site.

Sites B, C, D and E each occupy a single building: each needs a **LAN** (Local Area Network), built with Ethernet switches and Wi-Fi access points. Site A has two buildings 300 m apart on one campus: this is a **CAN** (Campus Area Network) — in practice two building LANs joined by a fibre optic backbone, since 300 m exceeds the 100 m limit of copper Ethernet. Inside each consultation room, a doctor's Bluetooth headset or a connected medical sensor would form a **PAN**, but this is optional here.

2. Classification of the whole.

The five sites spread over distances up to 130 km across the Region: together they form a **WAN** (Wide Area Network). More precisely, if CAMTEL fibre links the Bamenda–Ndop–Kumbo corridor, that portion behaves as a **MAN** (Metropolitan/regional Area Network); the whole interconnection, relying on operators' infrastructure (CAMTEL, MTN, Orange), is a WAN.

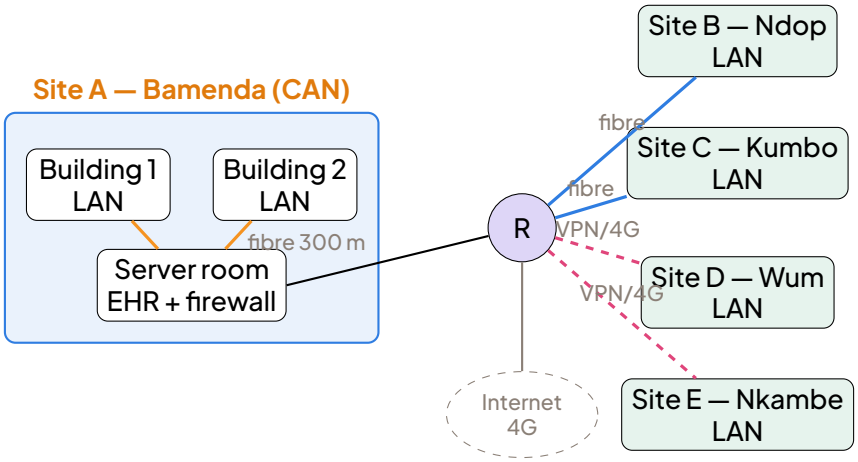
3. Choice of technologies per link.

Link	Technology	Justification
A ↔ B (40 km)	CAMTEL fibre / leased line	Fibre available on the corridor; EHR + video need bandwidth and reliability; cost acceptable for two hospitals
A ↔ C (110 km)	CAMTEL fibre / leased line	Same corridor; Kumbo has good coverage and 50 users
A ↔ D (80 km)	VPN over 4G	No fibre at Wum; 4G average but usable; VPN guarantees confidentiality; much cheaper than a leased line
A ↔ E (130 km)	VPN over 4G (VSAT as backup)	Only 8 users; 4G weak but sufficient for EHR access; VSAT too costly as main link, kept as backup option

4. The role of the VPN.

A **VPN** (Virtual Private Network) creates an encrypted tunnel between two sites across a public network such as the Internet or a 4G network. Data leaving site D is encrypted by the VPN gateway and only decrypted at site A; anyone intercepting the traffic on the operator's network reads only unintelligible ciphertext. It is indispensable here because medical records are confidential and the 4G network is a shared public infrastructure. The VPN protects the confidentiality and integrity of patient data in transit.

5. Logical diagram.



Logical design of the NWRHN: a CAN at Bamenda, LANs at each district site, fibre links (solid) on the CAMTEL corridor and encrypted VPN tunnels (dashed) over 4G for Wum and Nkambe.

6. Estimated budget (figures are indicative estimates in FCFA).

Item	Investment	Monthly
Site A: switches, cabling, fibre between buildings, server room firewall	6 000 000	—
Sites B, C: router + switch + firewall each	2 × 1 200 000	—
Sites D, E: 4G router with VPN + small switch each	2 × 600 000	—
CAMTEL leased lines (A, B, C)	—	3 × 150 000
4G business bundles (D, E)	—	2 × 40 000
Totals	**9 600 000**	**530 000**

7. Security controls. (a) Firewalls at every site filtering incoming traffic — against intrusions from the public network. (b) VPN encryption on 4G links — against interception of medical records. (c) User authentication with individual accounts and access rights on the EHR server — against unauthorised internal access; completed by regular backups of the EHR database against data loss.

8. Scalability. A future health centre only needs a 4G router and a VPN tunnel to site A: no new physical link is required, and the firewalls at site A just receive one more authorised tunnel. The design therefore grows at very low marginal cost.

EXERCISE 2 **Self-evaluation checklist**

Check your own design: (1) every site classified with a justification; (2) no fibre proposed where the brief says none exists; (3) VPN explained as encryption over a public network; (4) budget separates investment from recurrent costs; (5) at least three security controls, each matched to a threat; (6) one sentence on future growth.

7 METHODS AND WORKED EXAMPLES

7.1 Method 1 — Identifying the Type of Network from its Coverage Area

METHOD **Classifying a network from a description**

To classify a network described in an examination question:

1. Read the geographical clues: underline words such as a room, a building, a campus, a town, a country, the world.
2. Identify the owner: a single person (Bluetooth devices), one school, one company, a city council, a telecom operator (CAMTEL, MTN, Orange).
3. Match the scale to the class:
   - A few metres around one person → **PAN** (Personal Area Network), e.g. Bluetooth.
   - One room, building or small site (up to about 1–2 km) → **LAN** (Local Area Network), e.g. Ethernet/Wi-Fi in a school.
   - Several buildings of one institution within a few km → **CAN** (Campus Area Network).
   - A town or city (5–50 km) → **MAN** (Metropolitan Area Network), e.g. a city fibre ring.
   - A country, continent or the whole world → **WAN** (Wide Area Network), e.g. the Internet, a bank's national network.
4. Check the technology to confirm: PAN ≈ Bluetooth; LAN ≈ Ethernet, Wi-Fi; MAN ≈ fibre metro rings, WiMAX; WAN ≈ leased lines, fibre backbone, satellite, MPLS.
5. Write the answer with a justification: always name the class and quote the clue that proves it.

WORKED EXAMPLE 5 **Classifying four Cameroonian networks**

Classify each of the following networks as PAN, LAN, CAN, MAN or WAN, justifying your answer: (a) a student connects his Bluetooth earpiece to his smartphone; (b) the computer laboratory of GBHS Bamenda links 40 computers to one switch; (c) a university connects its faculties spread over three sites within Buea using its own fibre; (d) a bank connects its branches in Douala, Yaoundé and Garoua through leased lines from a telecom operator.

SOLUTION

Solution.

1. (a) PAN. The coverage is a few metres around one person and the technology is Bluetooth: this is the signature of a **Personal Area Network**.
2. (b) LAN. One room, one switch, 40 stations, Ethernet cabling: coverage limited to a single building ⇒ **Local Area Network**.
3. (c) CAN. Several buildings of the same institution (the university) interconnected over a few kilometres with its own fibre: larger than a LAN but still one organisation's campus ⇒ **Campus Area Network**.
4. (d) WAN. Branches in different towns, hundreds of kilometres apart, linked by leased lines rented from an operator: the organisation does not own the long-distance infrastructure ⇒ **Wide Area Network**.

Examiner's remark. In (c) the decisive clue is "three sites of the same institution within one town"; in (d) it is "different towns + operator's lines". Always quote the clue.

7.2 Method 2 — Distinguishing LAN, MAN and WAN in a Comparison Table

METHOD **Building a comparison table**

When asked to compare network classes:

1. Choose the comparison criteria: coverage, ownership, typical technology, speed, cost, example.
2. Fill one column per class (LAN, MAN, WAN), keeping each cell short and factual.
3. Remember the key contrasts: a LAN is owned by the organisation and is fast and cheap per user; a WAN rents transmission facilities from operators and is slower and more expensive per kilometre; a MAN sits in between (city scale, often run by a council or operator).
4. Conclude with one sentence stating which class suits which situation.

WORKED EXAMPLE 6 **Comparing LAN, MAN and WAN**

Draw a table comparing LAN, MAN and WAN according to: coverage area, ownership, typical technology, and one example each. Then state which type is appropriate for connecting the ministries located in the Yaoundé administrative quarter.

SOLUTION

Solution.

Criterion	LAN	MAN	WAN
Coverage	One building or small site (≤ 1–2 km)	A town or city (5–50 km)	Country, continent, world
Ownership	Single organisation	Council, operator or consortium	Operators; users lease lines
Technology	Ethernet, Wi-Fi	Metro fibre ring, WiMAX	Leased lines, fibre backbone, satellite, MPLS
Example	School computer lab	City fibre network of Douala	Internet; bank's national network

Choice for the ministries: the ministries of the Yaoundé administrative quarter are spread over a few kilometres within one city, so a **MAN** (a metropolitan fibre network) is appropriate: a single LAN cannot cover several separate buildings, while a WAN would be unnecessarily expensive since no inter-city links are needed.

7.3 Method 3 — Explaining Why the Internet is a WAN (Network of Networks)

METHOD **Justifying the status of the Internet**

To explain that the Internet is the largest WAN:

1. Recall the definition: a WAN interconnects LANs and MANs over long distances using infrastructure owned by telecommunication operators.
2. Show the chain: home/school LAN → Internet Service Provider (ISP) → national backbone → international links (submarine fibre cables, satellite).
3. Conclude: the Internet is a network of networks, worldwide in coverage, hence the largest possible WAN.
4. Mention the role of ISPs in Cameroon (CAMTEL, MTN, Orange) as the organisations whose WANs carry our traffic to the global Internet.

WORKED EXAMPLE 7 **From a school LAN to a website hosted abroad**

A student in a cybercafé in Bafoussam opens the website of a university hosted in Europe. Describe the path followed by the data across the different classes of networks, and de-

duce why the Internet is called "a network of networks".

SOLUTION

Solution.

1. The student's computer belongs to the cybercafé's LAN (Ethernet/Wi-Fi to the café's router).
2. The router forwards the request to the ISP's access network, then to the ISP's metropolitan and national infrastructure (MAN/WAN): e.g. the fibre backbone linking Bafoussam to Douala or Yaoundé.
3. The traffic reaches an international gateway and crosses a submarine fibre-optic cable (landing on the Cameroonian coast) and then European backbone networks (WAN).
4. Finally it enters the LAN of the data centre hosting the university's web server. The reply follows the reverse path.

Conclusion. The data crossed at least two LANs and several operators' WANs that are interconnected. The Internet is precisely this worldwide interconnection of millions of LANs, MANs and WANs that agree on common protocols (TCP/IP): it is therefore a **network of networks** and the largest WAN in existence.

7.4 Method 4 — Choosing the Right Network Type for an Organisation

METHOD **Selecting a network class for a given organisation**

To recommend a network type in a case study:

1. List the sites: how many buildings, in how many towns? Measure or estimate the distances between them.
2. Apply the scale rule: one site → LAN; several nearby sites of the same institution → CAN; several sites within one city → MAN; sites in different towns/countries → WAN (with a LAN in each site, interconnected by the WAN).
3. Consider ownership and budget: can the organisation lay its own cable (possible on a campus, impossible across a city without authorisation)? If not, lease capacity from an operator.
4. Consider needs: bandwidth (video, exams data, mobile money transactions), security (a bank encrypts its WAN links, e.g. with a VPN), reliability.
5. Write a justified recommendation: "I recommend a … because …".

WORKED EXAMPLE 8 **Recommending networks for three organisations**

For each organisation, recommend a network type (LAN, CAN, MAN or WAN) with two justifications: (a) a printing press occupying one building in Limbe; (b) a hospital complex with five pavilions on one fenced site; (c) a microfinance institution with a head office in Douala and eight branches in other towns of the Littoral and Centre regions.

SOLUTION

Solution.

1. (a) LAN. Justification 1: all users and machines are in a single building, within Ethernet/Wi-Fi range. Justification 2: a LAN is cheap, fast and fully owned and managed by the press; no operator link is needed.
2. (b) CAN. Justification 1: five separate buildings must be interconnected, which exceeds the reach of a single LAN. Justification 2: the pavilions stand on one fenced site, so the hospital can lay its own fibre between buildings (a campus network) without renting operator lines.
3. (c) WAN (with a LAN at each site). Justification 1: the branches are in different towns, tens to hundreds of kilometres apart, far beyond LAN/CAN/MAN range. Justification 2: the institution cannot own inter-city cables, so it must lease links (e.g. MPLS or VPN over the Internet) from an operator such as CAMTEL or MTN; because financial data are sensitive, the WAN traffic must be encrypted.

Key idea. In case (c) the complete solution is several LANs joined by a WAN: network classes combine; they do not exclude one another.

7.5 Method 5 — Estimating whether a Technology Can Cover a Given Distance

METHOD **Matching technology to distance and bandwidth**

1. Memorise the typical ranges: Bluetooth (≈ 10 m); Wi-Fi (≈ 30–100 m indoors); Ethernet on copper (≈ 100 m per segment); fibre on a campus (≈ a few km); metro fibre (≈ tens of km); operator backbone/satellite (≈ unlimited).
2. Compare the required distance with the range of each candidate technology.
3. Eliminate technologies that cannot reach, then among the survivors choose the cheapest that offers the required bandwidth.
4. If the distance exceeds what the organisation may cable itself (public roads, other towns), conclude that an operator's service is required ⇒ WAN logic.

WORKED EXAMPLE 9 **Choosing cabling technologies for a school**

A school has three blocks: the administration (A), the classrooms (B) 80 m from A, and the library (C) 600 m from A across the school yard. The school wants 1 Gbit/s between blocks. Propose a technology for each link, and state the resulting network class.

SOLUTION

Solution.

1. Link A–B (80 m): this is within the 100 m limit of copper Ethernet (twisted pair, Cat 5e/6), which carries 1 Gbit/s. A simple copper link or a Wi-Fi bridge is acceptable; copper is more reliable.
2. Link A–C (600 m): copper Ethernet cannot reach 600 m (limit 100 m) and Wi-Fi is unreliable at this distance outdoors. Optical fibre is required: it covers several kilometres at 1 Gbit/s and is immune to electromagnetic interference and lightning-induced surges.
3. Resulting class: three buildings of the same school on one site interconnected by its own cabling ⇒ a **CAN** (campus area network), itself made of three building LANs.

Examiner's remark. The trap is to answer "LAN" because it is one school: the multi-building structure with inter-building fibre makes it a campus network.

7.6 Method 6 — Drawing a Hierarchical Diagram of an Organisation's Network

METHOD **Representing LANs interconnected by a WAN**

1. Draw one cloud or box per site, labelled "LAN of …".
2. Draw a large cloud labelled with the WAN technology (e.g. "Operator MPLS / Internet + VPN").
3. Connect each site's router to the WAN cloud.
4. Annotate each link with its technology (fibre, 4G, leased line) and each site with its equipment (switch, router, firewall).
5. Keep the diagram readable: few words, clear labels.

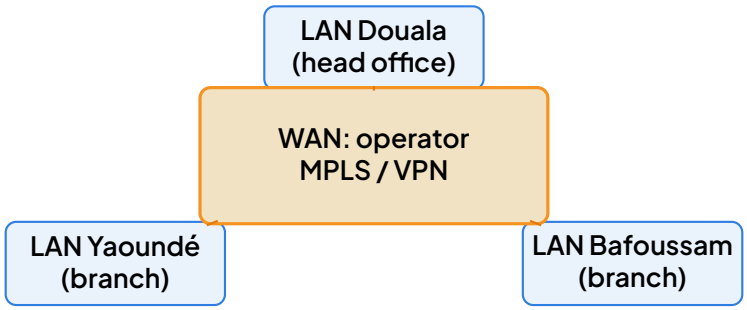
WORKED EXAMPLE 10 **Diagram for a company with three branches**

A company has its head office in Douala and branches in Yaoundé and Bafoussam. Each

site has a LAN with a switch and a router. Sketch the complete network and classify each part.

SOLUTION

Solution.



Three site LANs interconnected by an operator's WAN.

Classification. Each site network (computers → switch → router) is a **LAN**. The cloud interconnecting the three routers across different towns is a **WAN**, implemented by leasing a service (MPLS or encrypted VPN tunnels over the Internet) from a telecom operator. The company's complete network is therefore "three LANs joined by a WAN".

7.7 Method 7 — Integration: Designing the Complete Network of an Organisation (GCE-style)

METHOD **Solving a full network-design case study**

1. Analyse the situation: number of sites, distances, number of users, services needed (file sharing, Internet, video-surveillance, mobile money, e-mail).
2. Structure the solution in layers: a LAN per site; a CAN if several buildings of one site; a WAN to join distant sites; Internet access via an ISP.
3. Choose equipment: switches (LAN), routers (interconnection, WAN access), firewall (security), servers, Wi-Fi access points.
4. Choose WAN service: leased line/MPLS for guaranteed quality, or VPN over Internet for lower cost; justify with budget and security.
5. Justify every choice in one sentence; a justified choice earns marks, a bare list does not.

WORKED EXAMPLE 11 **Integration activity — the "Savannah College" project**

Savannah College has two campuses in Ngaoundéré (main campus: 3 buildings; annex campus: 1 building, 4 km away) and a small liaison office in Yaoundé. The college needs: internal file and printer sharing, a common student database hosted at the main campus, Internet access for all sites, and secure access for the accountant who travels. Design the network: classify each part, choose the interconnection technologies, and justify your choices.

SOLUTION

Solution.

1. LANs. Each of the 4 buildings (3 on the main campus, 1 annex) gets its own **LAN**: computers and printers connected to Ethernet switches, plus Wi-Fi access points for students. Justification: a building is within LAN range; Ethernet gives high speed at low cost.

2. CAN on the main campus. The 3 buildings of the main campus are interconnected by the college's own optical fibre ⇒ **CAN**. Justification: the college owns the site, so it can lay fibre; fibre carries the traffic of the student database server placed in the main building.

3. MAN between the two campuses. The annex is 4 km away, inside the same town. The college rents a metro fibre link from an operator (or uses a licensed radio bridge) ⇒ **MAN**. Justification: 4 km exceeds campus cabling rights (public roads), so an operator's metropolitan infrastructure is needed.

4. WAN to Yaoundé. The liaison office is in another city: the college leases a VPN over the Internet (cheaper) or an MPLS link (better guaranteed quality) ⇒ **WAN**. Justification: inter-city distance; only operators own such infrastructure.

5. Internet access and security. One router with a firewall at the main campus provides shared Internet access via an ISP (e.g. CAMTEL fibre); the travelling accountant connects through an encrypted VPN tunnel so that accounting data cannot be read on public networks.

Summary table.

Part	Class	Technology
Each building	LAN	Ethernet + Wi-Fi
Main campus (3 buildings)	CAN	College-owned fibre
Main campus ↔ annex	MAN	Operator metro fibre
Ngaoundéré ↔ Yaoundé	WAN	VPN/MPLS via operator

Conclusion. A real organisation's network is a combination of all four classes; the designer's skill is to choose, for each link, the cheapest technology that covers the distance with the required speed and security.

WARNING **Frequent mistakes at the GCE**

- Confusing Internet and WWW: the Internet is the physical WAN infrastructure; the Web is only one service running on it.
- Writing “WAN” for a large school: size in users does not matter, only geographical coverage and ownership of the links.
- Forgetting that a WAN solution always contains LANs at each site.
- Giving a network class without quoting the clue from the text that justifies it.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 3 **Multiple-choice questions**

For each question, choose the single correct answer.

1. A wireless link between a student's smartphone and her Bluetooth earpiece is an example of:

   (a) a PAN

   (b) a LAN

   (c) a MAN

   (d) a WAN

2. The computer network of Lycée de Nkol-Eton, covering all the buildings of the school campus, is best classified as:

   (a) a PAN

   (b) a LAN/CAN

   (c) a MAN

   (d) the Internet

3. A city-wide CCTV surveillance system interconnecting several police stations in Douala typically uses:

   (a) a PAN

   (b) a LAN

   (c) a MAN

   (d) a SAN

4. The national network interconnecting the branches of a bank in Yaoundé, Bafoussam, Garoua and Bamenda is:

   (a) a LAN

   (b) a CAN

   (c) a MAN

   (d) a WAN

EXERCISE 4 **True or false — justify**

State whether each statement is true or false, and justify your answer in one or two sentences.

1. A WAN is always owned by a single organisation.
2. The Internet is the largest example of a WAN.
3. A VPN allows a private network to be extended securely over a public network such as the Internet.
4. A MAN generally covers a smaller geographical area than a LAN.

EXERCISE 5 **Definitions**

Define each of the following terms in at most three lines, giving one concrete example for each:

1. Personal Area Network (PAN);
2. Local Area Network (LAN);
3. Metropolitan Area Network (MAN);
4. Wide Area Network (WAN);
5. Virtual Private Network (VPN).

EXERCISE 6 **Direct calculation — link cost**

An Internet Service Provider charges CFA 150 000 per kilometre per month for a dedicated fibre link.

1. Calculate the monthly cost of a 12 km link between two campuses of a university in Yaoundé.
2. Calculate the annual cost of this link.
3. The provider offers a 10 % discount for a 3-year commitment. Calculate the total amount paid over 3 years with the discount.

8.2 I practise

EXERCISE 7 **Matching scenarios to network types**

Match each scenario below to the most appropriate network type (PAN, LAN, CAN, MAN, WAN), justifying each choice in one sentence:

1. a home Bluetooth connection between a laptop and a printer;
2. the computer laboratory of Government High School Bafoussam;
3. a city-wide traffic-monitoring camera system in Yaoundé;
4. the interconnection of the regional offices of a microfinance institution across the ten regions of Cameroon;
5. a wireless sensor network on a 5 km^2 university campus in Dschang.

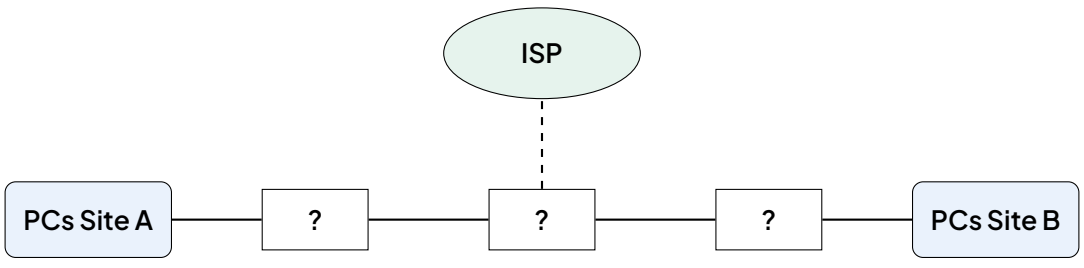
EXERCISE 8 **Advantages and disadvantages of a VPN**

A company with headquarters in Douala and a branch in Bafoussam hesitates between a leased line and a site-to-site IPsec VPN over the Internet.

1. List three advantages of the site-to-site IPsec VPN solution over the leased line.
2. List two disadvantages of the VPN solution.
3. Conclude by recommending one solution for a small enterprise with a limited budget, justifying your choice.

EXERCISE 9 **Completing a network diagram**

The diagram below shows the partial network of a company with two sites in the same town. Copy the diagram and complete it by inserting the appropriate device in each box (switch, router or firewall) and by labelling the network type of each segment (LAN or MAN).



Partial company network to be completed.

EXERCISE 10 **Cost comparison — leased line versus 4G backup**

A 10 Mbps MPLS link between Yaoundé and Douala (about 300 km) costs CFA 150 000 per kilometre per month. A 4G LTE backup connection costs CFA 25 000 per month with an availability of 99.5 %.

1. Calculate the monthly OPEX of the MPLS link.
2. Calculate the total cost of each solution over a 5-year horizon.
3. Compute, for the 4G solution, the expected number of hours of unavailability per month (take 1 month = 30 days).
4. Which solution is more cost-effective over 5 years? Discuss one non-financial factor that could still justify the MPLS link.

8.3 I prepare for the GCE

EXERCISE 11 **Structured question — classifying networks**

(a) Define the following network types and state the typical geographical coverage of each: PAN, LAN, MAN, WAN. (8 marks)
(b) For each of the following Cameroonian situations, identify the most appropriate network type and justify your answer:

1. a mobile money agent pairing his smartphone with a receipt printer;
2. the computer network of a hospital occupying a single building;
3. a fibre ring interconnecting the campuses of the University of Douala across the city;
4. the national backbone operated by CAMTEL interconnecting several towns. (8 marks)

(c) State two criteria, other than geographical coverage, that can be used to classify computer networks. (4 marks)

EXERCISE 12 **Case study essay — e-commerce expansion**

A Cameroonian e-commerce startup based in Yaoundé plans to expand to Douala and Bafoussam. Each new site will host about 20 employees, a local server room and point-of-sale terminals that must access the central database in Yaoundé in real time.
Write an essay of about one page in which you:

1. propose a network architecture for the three sites, indicating the network type (LAN, MAN, WAN) of each segment and the devices required (switches, routers, firewalls);

(8 marks)

2. justify the choice between a MAN, a WAN built on leased lines, and site-to-site VPNs over the Internet, considering cost, security, availability and scalability; (8 marks)
3. outline a four-step migration plan from the current single-site network to the new architecture, including staff training and a backup solution. (4 marks)

EXERCISE 13 **Problem — designing a school network**

The administration of a Government High School wants to interconnect three buildings (Administration, Classrooms block, Computer laboratory) spread over a campus of 800 m × 500 m, and to give teachers access to the Internet through a CAMTEL fibre subscription.
(a) State the type of network to be deployed and justify your answer. (2 marks)
(b) Draw a labelled diagram of the proposed network, showing the switch(es), the router, the firewall, the wireless access points and the ISP connection. (6 marks)
(c) The school hesitates between wired Ethernet and Wi-Fi for the classroom block. Give two advantages of each technology and make a justified recommendation. (6 marks)
(d) The ISP offers two subscriptions: 20 Mbps at CFA 75 000 per month and 50 Mbps at CFA 120 000 per month. The school has 200 students who each need at least 0.2 Mbps during practical sessions. Determine, by calculation, which subscription is sufficient, and compute its annual cost. (6 marks)

9 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple-choice questions**

1. Answer: a) a PAN. A Bluetooth link between a smartphone and an earpiece operates over a few metres around one person: this is the definition of a **Personal Area Network** (typical range ≤ 10 m).
2. Answer: b) a LAN/CAN. The network is confined to the buildings of a single school campus (a few hundred metres to about 1 km), which corresponds to a **Local Area Network**, more precisely a **Campus Area Network** when several buildings are interconnected.
3. Answer: c) a MAN. The system covers a whole city (Douala), i.e. a few kilometres to tens of kilometres: this is a **Metropolitan Area Network**.
4. Answer: d) a WAN. The branches are located in different towns spread over the national territory (hundreds of kilometres apart); interconnecting them requires a **Wide Area Network**, generally built on operator links (CAMTEL, MTN, Orange) or VPNs.

SOLUTION **Exercise 2 — True or false**

1. False. A WAN usually spans distances far beyond what one organisation can cable; it is generally built on leased infrastructure belonging to telecommunications operators (e.g. CAMTEL's backbone), or on VPN tunnels over the public Internet.
2. True. The Internet interconnects millions of networks on all continents; it is the largest WAN in existence — a "network of networks".
3. True. A **VPN** creates an encrypted tunnel through a public network (the Internet), so that remote sites or users communicate as if they were on the same private network, with confidentiality guaranteed by encryption (e.g. IPsec).
4. False. It is the opposite: a MAN covers a town or city (a few km to ≈ 100 km), whereas a LAN is limited to a building or a small site (a few metres to ≈ 1 km). The order of increasing size is PAN < LAN < MAN < WAN.

SOLUTION **Exercise 3 — Definitions**

1. PAN (Personal Area Network): a very small network interconnecting the personal de-

vices of one individual within a range of a few metres, usually wirelessly (Bluetooth, NFC). Example: a smartphone paired with a Bluetooth headset.

2. LAN (Local Area Network): a network covering a single building or a small site (up to about 1 km), owned and managed by one organisation, using Ethernet or Wi-Fi. Example: the computer laboratory network of a secondary school.
3. MAN (Metropolitan Area Network): a network covering a town or city (a few km to about 100 km), often built on fibre rings. Example: a fibre ring interconnecting the campuses of the University of Douala.
4. WAN (Wide Area Network): a network covering very large distances (regions, countries, continents), generally using operator infrastructure (leased lines, MPLS, satellite) or VPNs over the Internet. Example: the network interconnecting the branches of a bank across Cameroon.
5. VPN (Virtual Private Network): a technology that extends a private network securely over a public network by creating an encrypted tunnel between sites or users. Example: a site-to-site IPsec VPN linking the Douala headquarters of a company to its Bafoussam branch through the Internet.

SOLUTION **Exercise 4 — Direct calculation: link cost**

Given: unit price = CFA 150 000 per km per month; distance d = 12 km.

1. Monthly cost:
$$C_{\text{month}} = 150\,000 \times 12 = \text{CFA } 1\,800\,000.$$
2. Annual cost (12 months):
$$C_{\text{year}} = 1\,800\,000 \times 12 = \text{CFA } 21\,600\,000.$$
3. Cost over 3 years without discount: 21 600 000 × 3 = CFA 64 800 000.
With a 10 % discount:
$$C_{3\,\text{years}} = 64\,800\,000 \times (1 - 0.10) = 64\,800\,000 \times 0.9 = \text{CFA } 58\,320\,000.$$
The discount represents a saving of 64 800 000 − 58 320 000 = CFA 6 480 000.

SOLUTION **Exercise 5 — Matching scenarios to network types**

1. PAN — Bluetooth between a laptop and a printer works over a few metres around one user, which is the defining range of a PAN.

2. LAN — a computer laboratory is a single room/building owned by one school, the typical scope of a LAN.
3. MAN — the cameras are spread over the whole city of Yaoundé (several kilometres), which corresponds to a metropolitan coverage.
4. WAN — offices located in the ten regions of Cameroon are hundreds of kilometres apart, so only a WAN (operator links or VPNs) can interconnect them.
5. CAN (campus area network, a large LAN) — the sensors cover a single university campus of 5 km^2, i.e. several buildings of one site: larger than a room-scale LAN but far smaller than a MAN.

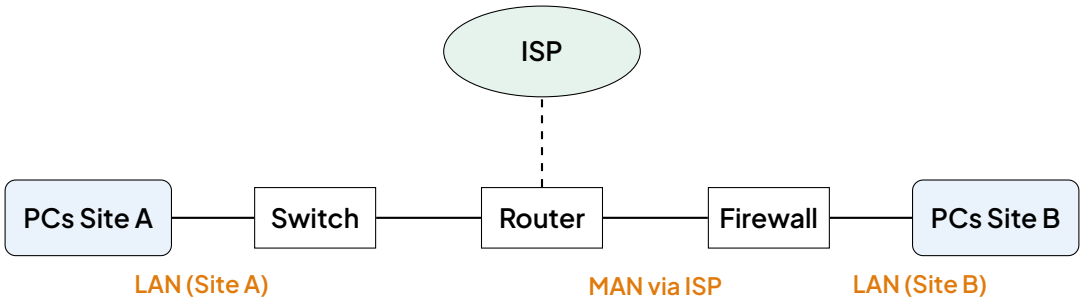
SOLUTION **Exercise 6 — Advantages and disadvantages of a VPN**

1. Three advantages of the site-to-site IPsec VPN over the leased line:
   - Lower cost: it reuses ordinary Internet subscriptions instead of paying an expensive dedicated link per kilometre.
   - Security by encryption: IPsec encrypts and authenticates all traffic in the tunnel, protecting the company's data on the public network.
   - Flexibility and scalability: adding a new site or a mobile worker only requires an Internet connection and a VPN configuration, with no new physical line to install.
2. Two disadvantages:
   - Performance depends on the public Internet: bandwidth and latency are not guaranteed (no service-level agreement on the tunnel itself), unlike a leased line.
   - Dependence on Internet availability and configuration complexity: if the ISP link fails the VPN fails too, and setting up IPsec correctly requires skilled staff (misconfiguration creates security holes).
3. Recommendation: for a small enterprise with a limited budget, the site-to-site IPsec VPN is recommended: it offers an acceptable level of security (encryption) at a fraction of the cost of a leased line, and it can be deployed quickly over existing Internet subscriptions. A 4G backup connection can be added to mitigate the availability weakness.

SOLUTION **Exercise 7 — Completing a network diagram**

Reasoning. Within each site, the PCs are interconnected by a switch: this segment is a LAN. To leave the site and reach the other site through the ISP, traffic must pass through

a router, which interconnects different networks; the link between the two sites via the ISP, within the same town, is a MAN. A firewall is placed at the boundary between the private network and the ISP to filter incoming and outgoing traffic. Hence, from left to right: switch, router, firewall.



Completed network: switch – router – firewall; LAN segments on each site, MAN link through the ISP.

Note: a symmetric drawing (firewall–router–switch on each side of the ISP cloud) is equally acceptable, provided each site has its own switch (LAN), a router to reach the other network, and a firewall protecting the connection to the ISP.

SOLUTION **Exercise 8 — Cost comparison: leased line versus 4G backup**

Given: MPLS: CFA 150 000/km/month over 300 km; 4G: CFA 25 000/month, availability 99.5 %.

1. Monthly OPEX of the MPLS link:

$$C_{\text{MPLS/month}} = 150\,000 \times 300 = \text{CFA } 45\,000\,000.$$

2. Over 5 years (= 60 months):

$C_{\text{MPLS}} = 45\,000\,000 \times 60 = \text{CFA } 2\,700\,000\,000$; $C_{4G} = 25\,000 \times 60 = \text{CFA } 1\,500\,000$.

3. Unavailability of the 4G solution: the unavailability rate is 100 % - 99.5 % = 0.5 %. One month = 30 × 24 = 720 h, hence:

$$T_{\text{down}} = 720 \times 0.005 = 3.6 \text{ h per month} \quad \text{(about 3 h 36 min)}.$$

4. Cost-effectiveness: the 4G solution is overwhelmingly cheaper (CFA 1.5 million versus CFA 2.7 billion over 5 years). Non-financial factor justifying the MPLS link: a leased MPLS line offers guaranteed bandwidth, low latency and a service-level agreement (committed availability, priority repair), whereas 4G bandwidth is shared and variable — for real-time applications (banking transactions, videoconferencing, a central database accessed in real time) this guaranteed quality of service can justify the cost.

SOLUTION **Exercise 9 — Structured question: classifying networks**

(a) (8 marks — 2 per network: 1 for the definition, 1 for the coverage)

- PAN: network interconnecting the personal devices of one user; coverage: a few metres (≤ 10 m), e.g. Bluetooth.
- LAN: network of a single building or small site, owned by one organisation; coverage: a few metres to about 1 km (Ethernet, Wi-Fi).
- MAN: network covering a town or city; coverage: a few km to about 100 km, usually fibre rings.
- WAN: network covering regions, countries or continents; coverage: hundreds to thousands of km, using operator infrastructure or VPNs over the Internet.

(b) (8 marks — 2 per situation: 1 for the type, 1 for the justification)

1. PAN — the smartphone and the receipt printer are paired by Bluetooth within arm's reach of the agent.
2. LAN — the hospital occupies a single building, so its network is local and owned by one organisation.
3. MAN — the fibre ring interconnects sites spread across the city of Douala, i.e. a metropolitan coverage.
4. WAN — CAMTEL's backbone interconnects towns hundreds of kilometres apart on the national territory.

(c) (4 marks — 2 per criterion, any two of)

- Transmission technology / media: wired (twisted pair, fibre) versus wireless (Wi-Fi, Bluetooth, 4G).
- Ownership / administration: private networks (owned by one organisation) versus public networks (operated by carriers).
- Topology: bus, star, ring, mesh. Architecture: client–server versus peer-to-peer.

Examiner's expectations: precise definitions with an order of magnitude for the coverage; justifications in (b) must explicitly refer to distance/coverage; in (c) the criteria must be other than coverage and must be named clearly.

SOLUTION **Exercise 10 — Case study essay: e-commerce expansion**

Indicative mark scheme (20 marks): (a) architecture 8; (b) justification 8; (c) migration plan 4. Award marks for a coherent structure, correct vocabulary (LAN/WAN/VPN, switch, router, firewall) and justified choices.
Model answer (outline of a full essay).

(a) Proposed architecture. Each of the three sites (Yaoundé headquarters, Douala and Bafoussam branches) deploys its own LAN: the 20 workstations, point-of-sale terminals and the local server are connected to Ethernet switches; Wi-Fi access points serve mobile devices. Each LAN is headed by a router that connects the site to the outside, protected by a firewall at the boundary with the Internet. Since the sites are in different towns, the interconnection is a WAN: site-to-site IPsec VPN tunnels over the Internet link Douala and Bafoussam to Yaoundé, where the central database server sits in a secured server room behind the firewall. The POS terminals therefore reach the database through the encrypted tunnel in real time.
(b) Choice of the interconnection. A MAN is impossible: the sites are in different cities, beyond metropolitan reach. A WAN built on leased lines offers guaranteed bandwidth, low latency and strong availability (SLA), but at a very high cost per kilometre per month — heavy for a startup — and each new site requires a new line. Site-to-site VPNs over the Internet are far cheaper (ordinary fibre/4G subscriptions), secure thanks to IPsec encryption and authentication, and highly scalable: a new branch only needs an Internet connection. Their weaknesses — variable performance and dependence on ISP availability — are mitigated by choosing good subscriptions and a 4G backup. Conclusion: VPN over the Internet is the best compromise of cost, security, availability and scalability for this startup.
(c) Four-step migration plan.

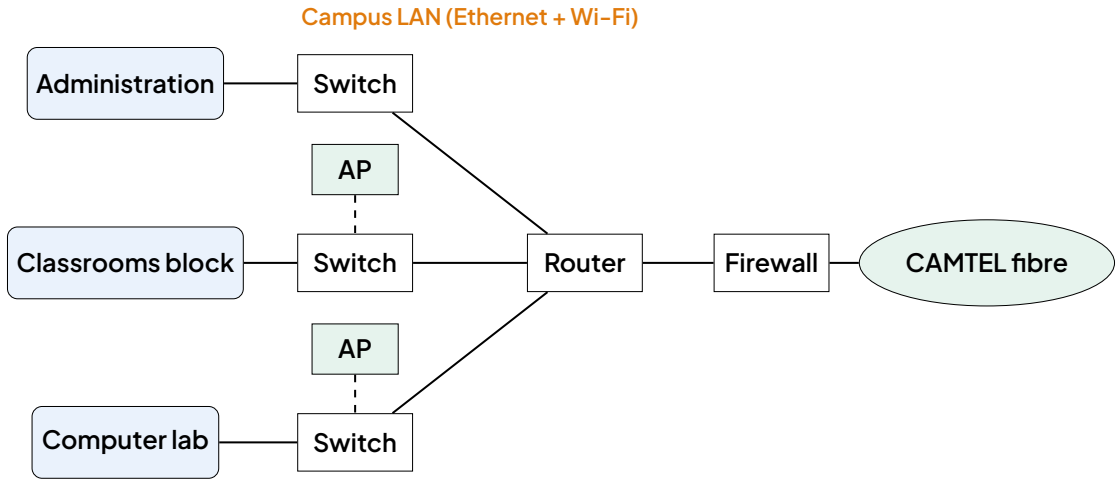
1. Audit and design: inventory the existing network, size bandwidth needs, plan IP addressing and the security policy.
2. Deploy and test: install the LANs, routers, firewalls and VPN tunnels at the new sites; test connectivity and real-time access to the database before opening.
3. Train staff: train employees on the new tools and security rules (passwords, VPN use), and train a local administrator per site.
4. Switch over and secure continuity: migrate progressively, set up a 4G backup link and regular backups of the central database (with copies kept off-site), then monitor the network.

Examiner's expectations: a structured essay with an introduction, labelled paragraphs (a), (b), (c), a comparison table or explicit criteria (cost, security, availability, scalability), and a justified conclusion.

SOLUTION **Exercise 11 — Problem: designing a school network**

(a) (2 marks) The network to deploy is a LAN, more precisely a Campus Area Network (CAN): the three buildings belong to a single school and are spread over a campus of 800 m × 500 m, i.e. within the local/campus range (well under a few kilometres), owned and managed by one organisation.
(b) (6 marks — 1 per correctly placed and labelled element: switches, router, firewall, ac-

cess points, ISP link, buildings)



Proposed school network: one switch per building, central router, firewall, Wi-Fi access points, CAMTEL fibre link.

(c) (6 marks — 2 advantages per technology, 2 for the justified recommendation)

- Wired Ethernet: (i) high, stable bandwidth and low latency, not affected by interference; (ii) better security, since an intruder must physically plug into the network.
- Wi-Fi: (i) mobility — students and teachers connect anywhere in the building with laptops and tablets; (ii) lower installation cost and no cabling work in existing classrooms, easy to extend.
- Recommendation: use wired Ethernet for the computer laboratory (fixed desktop PCs needing stable, fast connections for practical sessions) and Wi-Fi in the classroom block (mobile devices, flexibility), i.e. a hybrid solution combining the strengths of both.

(d) (6 marks — 2 for the need, 2 for the comparison, 2 for the annual cost)
Total bandwidth required during practical sessions:

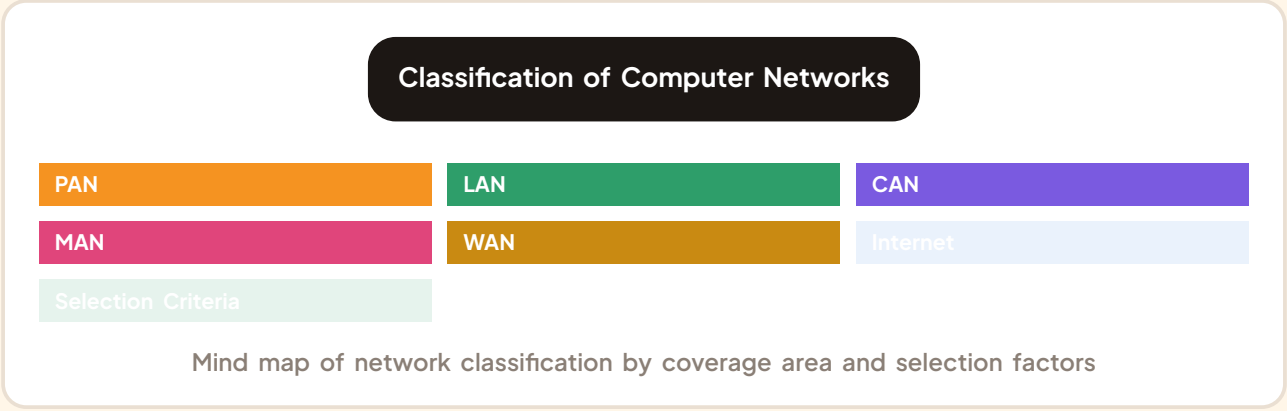
$$D = 200 \times 0.2 \text{ Mbps} = 40 \text{ Mbps}.$$

The 20 Mbps subscription ($20 < 40$) is insufficient; the 50 Mbps subscription ($50 \geq 40$) is sufficient.
Annual cost of the chosen subscription:

$$C = 120\,000 \times 12 = \text{CFA } 1\,440\,000 \text{ per year}.$$

Examiner's expectations: the diagram must be labelled and show the firewall between the router and the ISP; the calculation in (d) must compare the aggregated demand with each offer before concluding.

10 SUMMARY SHEET



Mind map of network classification by coverage area and selection factors

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **PAN** (Personal Area Network): covers a few metres, e.g. Bluetooth, USB, NFC.
- **LAN** (Local Area Network): covers a building or campus, high speed (100 Mbps–10 Gbps), owned by one organisation.
- **CAN** (Campus Area Network): interconnects several LANs within a limited geographical area (university, hospital).
- **MAN** (Metropolitan Area Network): spans a city, typically operated by a telecom provider (e.g. CAMTEL fibre ring).
- **WAN** (Wide Area Network): covers large distances (countries, continents), uses leased lines, satellite, MPLS, the Internet.
- **Internet**: global system of interconnected WANs using TCP/IP.
- **Bandwidth**: maximum data rate (bits/s) a link can carry.
- **Latency**: time for a packet to travel from source to destination.
- **Scalability**: ability to add users/devices without major redesign.
- **Security**: confidentiality, integrity, availability measures (firewalls, VPN, encryption).

Laws / Formulas / Theorems (examinable)

- No numeric formulae are required for this chapter. Focus on qualitative comparison tables and justification arguments.
- Remember the coverage–cost–performance trade-off: larger coverage → higher cost, lower bandwidth per user, higher latency.

Method: Choosing the Right Network for an Organisation

1. Define the geographical scope – single room (PAN), one building (LAN), multiple buildings (CAN), city (MAN), multi-city/country (WAN).
2. Estimate users and devices – count workstations, servers, IoT, mobile devices.
3. Determine bandwidth and latency requirements – video conferencing, cloud apps, real-time control.
4. Assess budget – capital expenditure (cabling, switches, routers) vs operational expenditure (leased line fees, ISP contracts).
5. Evaluate security and compliance – data sensitivity, regulatory needs (e.g. ANTIC guidelines), need for VPN or private MPLS.
6. Select technology and topology – Ethernet/Wi-Fi for LAN, fibre/MPLS for MAN/WAN, VPN over Internet for cost-effective secure WAN.
7. Plan for scalability and management – hierarchical design (core, distribution, access), monitoring tools, redundancy.

Common Mistakes (Traps)

- Confusing **CAN** with **MAN**: CAN is privately owned, limited to a campus; MAN is a public/operator service across a city.
- Thinking **WAN** always means "the Internet" – a private WAN (MPLS, leased line) is not the public Internet.
- Ignoring **latency** when choosing a WAN technology; satellite links have high latency unsuitable for real-time apps.
- Forgetting to justify the choice with at least three criteria (coverage, cost, bandwidth, security, scalability).
- Drawing network diagrams without labelling devices (router, switch, firewall) and media (fibre, copper, wireless).

GCE Advanced Level Tips

- Memorise the definition table (coverage, typical speed, ownership, example) for PAN, LAN, CAN, MAN, WAN, Internet.
- Be ready to compare two network types in a structured table (criteria as rows, network types as columns).
- In essay questions, state the scenario, list the selection criteria, then recommend with justification.
- Practice sketching a simple hierarchical network diagram (core–distribution–access) and label media types.

- Know the role of **ANTIC** (regulatory) and major Cameroonian ISPs (CAMTEL, MTN, Orange) as real-world examples.
- Time management: allocate 5 min for definitions, 10 min for comparison table, 15 min for design justification.

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